

Separation Laws for Flip Bifurcations and Endogenous Switching Boundaries in Gradient Maps

Zisong Yuan, Xiao Wen[†]

Department of Mathematics, Beihang University, Beijing, China

E-mail: alexsongyzs@gmail.com (Z. Yuan), wenxiao@buaa.edu.cn (X. Wen)

[†]Corresponding author.

Abstract

Gradient clipping is widely used to control large updates in optimization, while finite-step gradient descent develops period-two oscillations near the edge of linear stability. Recent work has analysed these oscillations via smooth flip bifurcation theory and higher-order stability criteria, whereas piecewise-smooth bifurcation theory describes generic interactions between period-doubling and switching boundaries. It remains unclear how a smooth gradient-descent oscillation first interacts with the switching surface generated endogenously by radial gradient clipping, and what dynamical structure emerges after this contact.

For the finite-dimensional clipped gradient map $F_{\eta,c}(x) = x - \eta \mathcal{C}_c^\circ(\nabla V(x))$, we analyse this transition near a nondegenerate strict local minimum with a simple maximal Hessian eigenvalue. Clipping is locally inactive near the minimum, so the fixed-point stability threshold and the local flip bifurcation coincide with those of ordinary gradient descent. Under a supercritical flip condition, the stable period-two branch reaches the clipping boundary at $\eta_{sc}(c) = \eta_f + K_\diamond c^2 + O(c^4)$ with $\eta_f = 2/\lambda_*$, where $K_\diamond = \Gamma_*/(3\lambda_*^4 \|v_*\|_\diamond^2)$ is determined explicitly by derivatives of the potential up to fourth order and by the clipping norm.

Beyond the separation law, the gradient structure yields rigidity absent from generic switching maps: the two points of every smooth period-two orbit have opposite gradients and hence reach a radial clipping boundary simultaneously; mixed smooth-clipped period-two itineraries are impossible; and every fully clipped period-two orbit has an exact unit Floquet multiplier and is therefore nonhyperbolic. A solvable planar model realizes the full transition from a stable smooth two-cycle to a neutral family of fully clipped period-two cycles.

Keywords: gradient maps; gradient clipping; flip bifurcation; period-doubling bifurcation; piecewise-smooth dynamical systems; switching hypersurfaces; border collision; periodic orbits.

1 Introduction

Gradient descent with a finite step size is naturally a discrete dynamical system,

$$x_{n+1} = x_n - \eta \nabla V(x_n), \quad (1.1)$$

and its behavior near the boundary of linear stability can differ substantially from that suggested by the continuous gradient-flow approximation; see, for example, [1, 2, 3, 4] for the classical optimization background. In particular, when a nondegenerate local minimum x_* has maximal Hessian eigenvalue

$$\lambda_* = \lambda_{\max}(D^2 V(x_*)),$$

the classical local stability threshold is

$$\eta_f = \frac{2}{\lambda_*}. \quad (1.2)$$

At this value, the multiplier corresponding to the sharpest eigendirection reaches -1 , and, under the usual nondegeneracy conditions, a flip bifurcation may occur. This is the classical simple flip-bifurcation

mechanism for smooth maps; see, for example, [5, 6, 7]. The dynamics of gradient descent near this threshold have recently received considerable attention, partly motivated by the edge-of-stability phenomenon observed in large-step neural-network optimization [8]. Beyond linear stability, higher-order derivatives of the objective determine whether the loss of stability gives rise to stable or unstable oscillations. In the multivariate setting, the nonlinear coefficient governing the local period-two branch depends not only on derivatives along the maximal-curvature eigendirection, but also on its coupling to the remaining Hessian eigendirections; see [9, 10, 11]. These developments provide a precise smooth bifurcation description of oscillatory gradient-descent dynamics near a minimum.

Gradient clipping is a standard mechanism for controlling large updates in optimization and has been studied from the viewpoints of training stability, adaptivity, and convergence [12, 13, 14, 15]. The situation changes qualitatively when the gradient is clipped. We consider the radial clipped gradient map

$$F_{\eta,c}(x) = x - \eta \mathcal{C}_c^\circ(\nabla V(x)), \quad (1.3)$$

where

$$\mathcal{C}_c^\circ(g) = \min \left\{ 1, \frac{c}{\|g\|_\diamond} \right\} g, \quad c > 0. \quad (1.4)$$

Here $\|\cdot\|_\diamond$ is a smooth norm away from the origin. The standard Euclidean clipping rule is recovered by taking $\|\cdot\|_\diamond = \|\cdot\|_2$.

The clipping rule introduces the state-dependent switching hypersurface

$$\Sigma_c = \{x : \|\nabla V(x)\|_\diamond = c\}. \quad (1.5)$$

Inside

$$\Omega_c^- = \{x : \|\nabla V(x)\|_\diamond < c\},$$

the map (1.3) coincides exactly with ordinary gradient descent, whereas in

$$\Omega_c^+ = \{x : \|\nabla V(x)\|_\diamond > c\}$$

the step magnitude is saturated. Thus clipping converts a smooth gradient map into a continuous piecewise-smooth dynamical system whose switching boundary is not imposed externally: it is generated endogenously by the same potential V that generates the smooth dynamics. Piecewise-smooth maps and their border-collision bifurcations have been studied extensively; see [16, 17, 18, 19, 20, 21, 22, 23].

This observation leads to a basic separation of mechanisms. Since

$$\nabla V(x_*) = 0,$$

for every fixed $c > 0$ there exists a neighborhood of x_* entirely contained in Ω_c^- . Consequently, clipping cannot change the local germ of the map at the minimum. In particular, the fixed point, its linearization, its first loss-of-stability threshold, the center-manifold reduction, and the smooth flip coefficient are all identical to those of ordinary gradient descent. The first genuinely clipping-induced effect must therefore occur at finite amplitude, after the bifurcating orbit has grown far enough to reach Σ_c .

The aim of this paper is to characterize this first smooth-to-switching transition.

1.1 Main question

Assume that x_* is a nondegenerate strict local minimum, that the maximal Hessian eigenvalue λ_* is simple, and that the ordinary gradient map undergoes a supercritical flip bifurcation at

$$\eta_f = \frac{2}{\lambda_*}.$$

For

$$\mu = \eta - \eta_f > 0$$

small, the smooth map possesses a stable period-two cycle

$$\mathcal{O}_2(\mu) = \{x_-(\mu), x_+(\mu)\}. \quad (1.6)$$

The central question is:

At what parameter value does the smooth period-two branch first reach the clipping boundary, and what dynamical structure is created when this contact occurs?

For a generic flip bifurcation, the oscillation amplitude grows on the square-root scale

$$\|x_{\pm}(\mu) - x_*\| \asymp \sqrt{\mu},$$

whereas the local switching boundary generated by

$$\|\nabla V(x)\|_{\diamond} = c$$

lies at distance $O(c)$ from the critical point. This suggests the matching law

$$\mu_{\text{sc}} \asymp c^2.$$

However, the main issue is not merely the exponent. General period-doubling–border-collision interactions are already known to produce tangencies between smooth bifurcation and switching loci. What is special here is the gradient-generated nature of both the smooth map and the switching surface.

This additional structure allows us to derive the separation coefficient explicitly from the local derivatives of V , to prove that the two points of the period-two orbit reach the switching surface simultaneously, and to determine a universal spectral degeneracy in the fully clipped regime.

1.2 Main results

Let

$$H_* = D^2V(x_*), \quad T_* = D^3V(x_*), \quad Q_* = D^4V(x_*),$$

and let

$$H_*v_* = \lambda_*v_*, \quad \|v_*\|_2 = 1.$$

Define

$$g_* = T_*[v_*, v_*, \cdot]^{\sharp} \tag{1.7}$$

and

$$\Gamma_* = 3T_*[v_*, v_*, H_*^{-1}g_*] - Q_*[v_*, v_*, v_*, v_*]. \tag{1.8}$$

The condition

$$\Gamma_* > 0$$

is the supercritical flip condition used throughout the paper.

Our first result is a local identity principle. For every fixed $c > 0$, the clipped and unclipped maps coincide on a neighborhood of x_* . Hence

$$DF_{\eta,c}(x_*) = I - \eta H_*, \tag{1.9}$$

and the first local stability threshold is exactly

$$\boxed{\eta_f = \frac{2}{\lambda_*}}, \tag{1.10}$$

independently of c . More strongly, the finite jets of the two maps at x_* agree, so the local center dynamics and smooth flip coefficient are also clipping-independent.

Second, we derive the leading amplitude of the smooth period-two branch. For

$$\mu = \eta - \eta_f \rightarrow 0^+,$$

one has

$$x_{\pm}(\mu) = x_* \pm A_*\sqrt{\mu}v_* + O(\mu), \tag{1.11}$$

where

$$\boxed{A_*^2 = \frac{3\lambda_*^2}{\Gamma_*}}. \tag{1.12}$$

Third, the gradient structure gives an exact identity that is stronger than the local normal-form expansion. For every genuine period-two cycle of ordinary gradient descent,

$$\boxed{\nabla V(x_+) = -\nabla V(x_-).} \quad (1.13)$$

Thus the two points have exactly the same clipping exposure for every norm:

$$\|\nabla V(x_+)\|_\diamond = \|\nabla V(x_-)\|_\diamond.$$

Consequently, the first interaction with a radial clipping boundary is necessarily a simultaneous two-point contact.

Along the bifurcating branch, the common exposure satisfies

$$\|\nabla V(x_\pm(\mu))\|_\diamond = B_*\sqrt{\mu} + O(\mu^{3/2}), \quad (1.14)$$

where

$$B_* = \lambda_* A_* \|v_*\|_\diamond. \quad (1.15)$$

Inverting the contact equation

$$\|\nabla V(x_\pm(\mu_{\text{sc}}))\|_\diamond = c$$

gives our principal separation theorem:

$$\boxed{\eta_{\text{sc}}(c) - \eta_f = K_\diamond c^2 + O(c^4), \quad c \rightarrow 0^+,} \quad (1.16)$$

with the explicit coefficient

$$\boxed{K_\diamond = \frac{\Gamma_*}{3\lambda_*^4 \|v_*\|_\diamond^2}.} \quad (1.17)$$

Equivalently, if

$$\{v_i\}_{i=1}^d$$

is an orthonormal Hessian eigenbasis with eigenvalues λ_i , then

$$K_\diamond = \frac{3 \sum_{i=1}^d \frac{T_*[v_*, v_*, v_i]^2}{\lambda_i} - Q_*[v_*^4]}{3\lambda_*^4 \|v_*\|_\diamond^2}. \quad (1.18)$$

Thus the same high-order local quantity that controls the birth and stability of the smooth oscillation also determines the leading distance in parameter space to the first clipping event.

The radial structure imposes further restrictions after contact. A period-two orbit of the clipped map cannot have one point in Ω_c^- and the other in Ω_c^+ . Hence a mixed smooth-clipped period-two itinerary is impossible. If a period-two orbit persists beyond contact, both points must lie in the clipped region and satisfy the exact amplitude constraint

$$\boxed{\|y_+ - y_-\|_\diamond = \eta c.} \quad (1.19)$$

Finally, every fully clipped period-two orbit is necessarily nonhyperbolic. If

$$M_c = DF_{\eta,c}(y_-)DF_{\eta,c}(y_+)$$

denotes its monodromy matrix, then

$$\boxed{1 \in \sigma(M_c).} \quad (1.20)$$

This unit multiplier follows from the fact that radial clipping annihilates infinitesimal changes in gradient magnitude while retaining only tangent variations of the gradient direction.

A completely solvable planar gradient model shows that this degeneracy can have a genuine nonlinear manifestation: after the simultaneous switching contact, the isolated attracting smooth period-two cycle is replaced locally by a one-parameter family of fully clipped period-two cycles, transversely attracting but neutral along the family.

1.3 Relation to existing bifurcation theory

The present analysis lies at the intersection of two established lines of work.

The first is smooth local bifurcation theory for maps. Near a simple multiplier -1 , center-manifold and normal-form reductions yield the classical flip bifurcation and the square-root scaling of the emerging period-two branch; see, for example, [5, 6, 24, 25, 26, 27, 7, 28].

More recently, nonlinear oscillatory dynamics of gradient descent near the edge of stability have been studied directly. Mulayoff and Stich [9] derived a multivariate stability criterion for nonlinear gradient-descent oscillations near a minimum in terms of higher-order derivatives of the objective. Gan [10] formulated edge-of-stability dynamics in a bifurcation-theoretic framework, in which the relevant normal dynamics undergo a flip bifurcation governed by the corresponding first Lyapunov coefficient. Relatedly, Litman [11] characterized fixed points and period-two orbits of gradient descent through an exact two-step variational formulation and obtained a local reduction in terms of the half-amplitude of the oscillation.

Some ingredients of the smooth analysis developed below therefore belong to standard flip-bifurcation theory or have close analogues in recent analyses of nonlinear gradient-descent dynamics. We nevertheless include self-contained derivations, both to fix notation and because the resulting coefficients enter explicitly into the subsequent switching analysis.

The second relevant line of work is the theory of piecewise-smooth maps and border-collision bifurcations [29, 30, 31, 18, 21, 22]. In particular, Simpson and Meiss [32, 33, 34, 35, 36] studied the codimension-two interaction between period-doubling and border-collision bifurcations in general piecewise-smooth continuous maps. They showed that the border-collision locus of the period-doubled solution can be tangent to the period-doubling locus near the codimension-two point. Thus quadratic separation between smooth bifurcation and switching loci is already a feature of generic period-doubling–border-collision interactions.

Our contribution is not the generic existence of such a tangency, nor the smooth flip mechanism itself. Rather, we exploit the additional variational structure of clipped gradient maps. The switching surface

$$\Sigma_c = \{x : \|\nabla V(x)\|_\diamond = c\}$$

is generated endogenously by the same potential that determines the smooth gradient dynamics. Moreover, the period-two equations impose exact gradient-balance identities.

These additional constraints yield structural consequences that are absent from a generic externally switched map. In particular, we obtain:

1. clipping-independent local flip dynamics;
2. an explicit tensor formula for the flip-to-switching separation coefficient;
3. simultaneous switching contact of the two points of a smooth period-two orbit;
4. nonexistence of mixed smooth–clipped period-two itineraries;
5. an exact unit Floquet multiplier for every fully clipped period-two orbit.

It is these gradient-induced rigidity properties, rather than the standard smooth flip bifurcation or the generic tangency mechanism, that form the main focus of the present work.

1.4 Organization of the paper

The paper is organized as follows.

Section 2 introduces the clipped gradient map, the switching hypersurface, and the standing assumptions.

Section 3 proves the local identity between clipped and ordinary gradient descent and establishes the clipping-independent stability threshold.

Section 4 derives the supercritical period-two branch and its explicit leading amplitude.

Section 5 develops the exact gradient balance and switching geometry along the period-two branch.

Section 6 proves the quadratic flip-to-switching separation law and derives its explicit coefficient.

Section 7 studies the fully clipped regime and proves the universal unit-multiplier theorem.

Section 8 presents a completely solvable planar model that realizes the full transition analytically.

2 Problem setting and standing hypotheses

Throughout the paper, $\langle \cdot, \cdot \rangle$ and $\| \cdot \|_2$ denote the standard Euclidean inner product and norm on $\mathbb{R}^d$, respectively. All gradients, Hessians, and spectral quantities are understood with respect to this Euclidean structure.

Let $\mathcal{U} \subset \mathbb{R}^d$ be an open set and let

$$V : \mathcal{U} \rightarrow \mathbb{R}$$

be a smooth potential. For a step size $\eta > 0$, the associated gradient-descent map is

$$G_\eta(x) = x - \eta \nabla V(x). \quad (2.1)$$

Our aim is to study how a smooth flip bifurcation of (2.1) interacts with the switching boundary generated intrinsically by gradient clipping.

2.1 Clipped gradient maps

Let $\| \cdot \|_\diamond$ be a norm on $\mathbb{R}^d$. For a clipping threshold $c > 0$, define

$$\mathcal{C}_c^\diamond(g) = \begin{cases} g, & \|g\|_\diamond \leq c, \\ \frac{c}{\|g\|_\diamond} g, & \|g\|_\diamond > c, \end{cases} \quad g \in \mathbb{R}^d. \quad (2.2)$$

Equivalently,

$$\mathcal{C}_c^\diamond(g) = \min \left\{ 1, \frac{c}{\|g\|_\diamond} \right\} g,$$

with the convention $\mathcal{C}_c^\diamond(0) = 0$.

The clipped gradient map is

$$F_{\eta,c}(x) = x - \eta \mathcal{C}_c^\diamond(\nabla V(x)). \quad (2.3)$$

The standard Euclidean gradient-clipping rule corresponds to

$$\| \cdot \|_\diamond = \| \cdot \|_2.$$

Associated with (2.3) is the endogenous switching hypersurface

$$\Sigma_c = \{x \in \mathcal{U} : \|\nabla V(x)\|_\diamond = c\}. \quad (2.4)$$

It separates the local phase space into the smooth region

$$\Omega_c^- = \{x \in \mathcal{U} : \|\nabla V(x)\|_\diamond < c\} \quad (2.5)$$

and the clipped region

$$\Omega_c^+ = \{x \in \mathcal{U} : \|\nabla V(x)\|_\diamond > c\}. \quad (2.6)$$

Accordingly,

$$F_{\eta,c}(x) = G_\eta(x) = x - \eta \nabla V(x), \quad x \in \Omega_c^-, \quad (2.7)$$

whereas in Ω_c^+ ,

$$F_{\eta,c}(x) = x - \eta c \frac{\nabla V(x)}{\|\nabla V(x)\|_\diamond}. \quad (2.8)$$

Thus $F_{\eta,c}$ is continuous and piecewise smooth, while its switching geometry is not prescribed independently of the dynamics: the boundary Σ_c is generated by the same potential V that defines the gradient map.

2.2 The distinguished critical point

We focus on the dynamics near an isolated nondegenerate local minimum x_* of V .

Set

$$H_* := D^2V(x_*). \quad (2.9)$$

Since H_* is symmetric, choose an orthonormal eigenbasis $\{v_i\}_{i=1}^d$ satisfying

$$H_*v_i = \lambda_i v_i, \quad 0 < \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_d. \quad (2.10)$$

We denote the maximal eigenvalue and its corresponding unit eigenvector by

$$\lambda_* := \lambda_d, \quad v_* := v_d. \quad (2.11)$$

The linearization of the smooth gradient map at x_* is

$$DG_\eta(x_*) = I - \eta H_*, \quad (2.12)$$

and hence its multipliers are

$$\rho_i(\eta) = 1 - \eta \lambda_i, \quad i = 1, \dots, d. \quad (2.13)$$

The distinguished step size at which the maximal-curvature multiplier reaches -1 is

$$\eta_f := \frac{2}{\lambda_*}. \quad (2.14)$$

At $\eta = \eta_f$,

$$\rho_d(\eta_f) = -1. \quad (2.15)$$

The parameter

$$\mu := \eta - \eta_f \quad (2.16)$$

will be used as the local bifurcation parameter throughout the paper.

2.3 Higher-order tensors and the flip coefficient

We write

$$T_* := D^3V(x_*), \quad Q_* := D^4V(x_*), \quad (2.17)$$

viewed as symmetric multilinear forms.

For $u, w \in \mathbb{R}^d$, let

$$T_*[u, w, \cdot]^\sharp \in \mathbb{R}^d \quad (2.18)$$

denote the Euclidean Riesz representative of the linear functional $z \mapsto T_*[u, w, z]$; namely,

$$\langle T_*[u, w, \cdot]^\sharp, z \rangle = T_*[u, w, z] \quad \text{for all } z \in \mathbb{R}^d. \quad (2.19)$$

Define

$$g_* := T_*[v_*, v_*, \cdot]^\sharp \quad (2.20)$$

and

$$h_* := H_*^{-1}g_*. \quad (2.21)$$

The nonlinear quantity governing the local flip bifurcation is

$$\Gamma_* := 3T_*[v_*, v_*, h_*] - Q_*[v_*, v_*, v_*]. \quad (2.22)$$

Equivalently, using the eigenbasis (2.10),

$$\Gamma_* = 3 \sum_{i=1}^d \frac{T_*[v_*, v_*, v_i]^2}{\lambda_i} - Q_*[v_*, v_*, v_*]. \quad (2.23)$$

With the flip-normal-form convention used in this paper, define

$$\ell_* := \frac{\eta_f}{6} \Gamma_* = \frac{\Gamma_*}{3\lambda_*}. \quad (2.24)$$

Thus $\ell_* > 0$ corresponds to the supercritical case in which a stable small-amplitude period-two branch is created for $\eta > \eta_f$.

The expression (2.23) is the multivariate gradient-map form of the nonlinear flip criterion: the cubic coupling of the critical direction to all Hessian eigendirections contributes to the sign of the bifurcation coefficient, rather than only the derivatives taken purely along v_* .

2.4 Standing hypotheses

The following assumptions are imposed unless stated otherwise.

(H1) Regularity. The potential satisfies

$$V \in C^7(\mathcal{U}, \mathbb{R}).$$

(H2) Nondegenerate strict local minimum. The distinguished point $x_* \in \mathcal{U}$ satisfies

$$\nabla V(x_*) = 0, \quad H_* = D^2V(x_*) \succ 0. \quad (2.25)$$

(H3) Simple maximal curvature. The largest Hessian eigenvalue is simple:

$$\lambda_{d-1} < \lambda_d = \lambda_*. \quad (2.26)$$

(H4) Supercritical flip nondegeneracy. The coefficient defined in (2.22) satisfies

$$\Gamma_* > 0, \quad (2.27)$$

or equivalently

$$\ell_* > 0.$$

(H5) Smooth clipping geometry. The clipping norm $\|\cdot\|_\diamond$ is of class C^3 on $\mathbb{R}^d \setminus \{0\}$.

2.5 Immediate spectral consequences

Assumptions **(H2)**–**(H3)** imply that the spectral hypotheses required for a simple codimension-one flip bifurcation are automatic for the gradient map.

Indeed, for $i = 1, \dots, d-1$,

$$0 < \lambda_i < \lambda_*,$$

and therefore, at $\eta = \eta_f$,

$$-1 < 1 - \frac{2\lambda_i}{\lambda_*} < 1. \quad (2.28)$$

Hence -1 is the unique multiplier of $DG_{\eta_f}(x_*)$ on the unit circle.

Moreover,

$$\frac{d}{d\eta}\rho_d(\eta) = -\lambda_*, \quad (2.29)$$

so that

$$\left. \frac{d}{d\eta}\rho_d(\eta) \right|_{\eta=\eta_f} = -\lambda_* \neq 0. \quad (2.30)$$

Thus the critical multiplier crosses -1 transversally as η passes through η_f .

Thus, under the spectral assumptions **(H2)**–**(H3)**, the sign of Γ_* provides the remaining nonlinear nondegeneracy condition that distinguishes the supercritical and subcritical flip cases.

2.6 Local form of the switching geometry

We record one elementary observation that motivates the later switching analysis.

Let

$$x = x_* + sv_* + r(s), \quad r(s) = O(s^2), \quad s \rightarrow 0.$$

Since $\nabla V(x_*) = 0$ and $H_*v_* = \lambda_*v_*$, Taylor expansion gives

$$\nabla V(x) = \lambda_*sv_* + O(s^2). \quad (2.31)$$

By positive homogeneity and smoothness of the clipping norm away from the origin,

$$\|\nabla V(x)\|_\diamond = \lambda_*\|v_*\|_\diamond|s| + O(s^2). \quad (2.32)$$

In particular, the leading-order displacement of the switching boundary from x_* in the critical direction is nondegenerate.

For the Euclidean clipping rule, $\|v_*\|_2 = 1$, and (2.32) reduces to

$$\|\nabla V(x)\|_2 = \lambda_*|s| + O(s^2). \quad (2.33)$$

A precise expansion of Σ_c along the center manifold will be derived later in Section 5.

2.7 Scope of the present paper

The positive-definiteness assumption in **(H2)** deliberately restricts the present work to an isolated strict local minimum. This excludes the additional unit multiplier $+1$ that occurs when the Hessian has a nontrivial kernel, as in Morse–Bott manifolds of minimizers.

The purpose of this restriction is to isolate the interaction between a simple smooth flip bifurcation and the endogenous switching boundary generated by clipping. Extensions involving manifolds of minimizers, multiple critical multipliers, or degenerate flip coefficients require a higher-dimensional center dynamics and are outside the scope of the present paper.

3 Local structure and clipping-independent stability

We first establish a structural fact used throughout the paper: for every fixed $c > 0$, clipping is completely inactive in a sufficiently small neighborhood of x_* , so the clipped and unclipped gradient maps have identical local germs at x_* . Consequently, clipping cannot alter the linear stability threshold, the local invariant structures, or any smooth local normal-form coefficient at x_* ; its influence can appear only when an orbit reaches a finite-amplitude region intersecting the switching hypersurface Σ_c .

3.1 Fixed points

We begin with a global observation.

Proposition 3.1 (Fixed-point invariance). *Let $\eta > 0$ and $c > 0$. Then*

$$\text{Fix}(F_{\eta,c}) = \text{Crit}(V) := \{x \in \mathcal{U} : \nabla V(x) = 0\}. \quad (3.1)$$

In particular, the set of fixed points is independent of both η and c .

Proof. If $\nabla V(x) = 0$, then

$$\mathcal{C}_c^\diamond(\nabla V(x)) = 0,$$

and hence

$$F_{\eta,c}(x) = x - \eta \mathcal{C}_c^\diamond(\nabla V(x)) = x.$$

Thus every critical point of V is a fixed point of $F_{\eta,c}$.

Conversely, suppose that

$$F_{\eta,c}(x) = x.$$

Since $\eta > 0$, we obtain

$$\mathcal{C}_c^\diamond(\nabla V(x)) = 0.$$

By the definition of the radial clipping operator (2.2),

$$\mathcal{C}_c^\diamond(g) = 0 \iff g = 0.$$

Therefore

$$\nabla V(x) = 0.$$

This proves (3.1). □

Remark 3.2. Proposition 3.1 shows that norm clipping does not create spurious equilibria. All dynamical effects of clipping therefore arise from the behavior of nonstationary trajectories rather than from a modification of the critical-point set of the potential.

3.2 Quantitative inactivity of clipping

The next lemma gives a quantitative local estimate.

For a linear map $A : \mathbb{R}^d \rightarrow \mathbb{R}^d$, define the mixed operator norm

$$\|A\|_{2 \rightarrow \diamond} := \sup_{\|z\|_2=1} \|Az\|_\diamond. \quad (3.2)$$

Lemma 3.3 (Local gradient estimate). *Under (H1)–(H2), there exist $r_0 > 0$ and $L_* > 0$ such that*

$$\overline{B_{r_0}(x_*)} \subset \mathcal{U}$$

and

$$\|\nabla V(x)\|_\diamond \leq L_* \|x - x_*\|_2 \quad \text{for all } x \in B_{r_0}(x_*). \quad (3.3)$$

Proof. Since $\mathcal{U}$ is open and $x_* \in \mathcal{U}$, choose $r_0 > 0$ sufficiently small so that

$$\overline{B_{r_0}(x_*)} \subset \mathcal{U}.$$

By (H1), the Hessian D^2V is continuous. Hence

$$L_* := \sup_{y \in \overline{B_{r_0}(x_*)}} \|D^2V(y)\|_{2 \rightarrow \diamond} < \infty. \quad (3.4)$$

For $x \in B_{r_0}(x_*)$, the line segment joining x_* to x lies in $B_{r_0}(x_*)$. Since $\nabla V(x_*) = 0$, the fundamental theorem of calculus gives

$$\begin{aligned} \nabla V(x) &= \nabla V(x) - \nabla V(x_*) \\ &= \int_0^1 D^2V(x_* + t(x - x_*))(x - x_*) dt. \end{aligned} \quad (3.5)$$

Therefore

$$\begin{aligned} \|\nabla V(x)\|_\diamond &\leq \int_0^1 \|D^2V(x_* + t(x - x_*))\|_{2 \rightarrow \diamond} \|x - x_*\|_2 dt \\ &\leq L_* \|x - x_*\|_2. \end{aligned} \quad (3.6)$$

This proves (3.3). $\square$

For every fixed $c > 0$, define

$$r_c := \min \left\{ r_0, \frac{c}{2L_*} \right\}. \quad (3.7)$$

Corollary 3.4 (Clipping inactivity). *For every $c > 0$,*

$$B_{r_c}(x_*) \subset \Omega_c^-. \quad (3.8)$$

Equivalently,

$$B_{r_c}(x_*) \cap \Sigma_c = \emptyset. \quad (3.9)$$

Proof. If $x \in B_{r_c}(x_*)$, then by Lemma 3.3,

$$\|\nabla V(x)\|_\diamond \leq L_* \|x - x_*\|_2 < L_* r_c \leq \frac{c}{2} < c.$$

Hence $x \in \Omega_c^-$. $\square$

Remark 3.5 (Dependence on the clipping threshold). The neighborhood in Corollary 3.4 is not uniform as $c \rightarrow 0^+$. The explicit choice (3.7) has

$$r_c = O(c).$$

This shrinking local scale is essential for the later smooth-to-switching analysis: although clipping is invisible in a neighborhood of x_* for every fixed $c > 0$, the boundary of that neighborhood approaches the critical point as $c \rightarrow 0^+$.

3.3 Local identity of the clipped and smooth maps

We can now state the basic structural theorem.

Theorem 3.6 (Local identity theorem). *Assume (H1)–(H2). For every fixed $c > 0$, there exists $r_c > 0$ such that, for every $\eta > 0$,*

$$F_{\eta,c}(x) = G_\eta(x) = x - \eta \nabla V(x) \quad \text{for all } x \in B_{r_c}(x_*). \quad (3.10)$$

The radius r_c may be chosen independently of η .

Proof. By Corollary 3.4,

$$B_{r_c}(x_*) \subset \Omega_c^-.$$

Hence, for every $x \in B_{r_c}(x_*)$,

$$\|\nabla V(x)\|_\diamond < c.$$

By definition of the clipping operator,

$$\mathcal{C}_c^\diamond(\nabla V(x)) = \nabla V(x).$$

Substitution into (2.3) yields

$$F_{\eta,c}(x) = x - \eta \nabla V(x) = G_\eta(x).$$

The construction of r_c in (3.7) does not involve η , so the same neighborhood works for every $\eta > 0$. $\square$

Theorem 3.6 is stronger than equality of the linearizations: the two maps agree as functions on an open neighborhood of x_* .

3.4 Jet invariance and local normal forms

The local identity immediately implies equality of all derivatives allowed by the regularity of V .

Corollary 3.7 (Local jet and normal-form invariance). *Under (H1)–(H4), for every fixed $c > 0$ and $\eta > 0$:*

- (i) *every local Taylor coefficient of $F_{\eta,c}$ and G_η at x_* that is defined under the assumed regularity is identical; in particular,*

$$DF_{\eta,c}(x_*) = I - \eta H_*; \quad (3.11)$$

- (ii) *the local center-manifold equations and all smooth flip normal-form coefficients of $F_{\eta,c}$ at x_* are identical to those of G_η ; in particular, the coefficient ℓ_* defined in (2.24) is independent of the clipping threshold c .*

Proof. The two maps coincide on a neighborhood of x_* by Theorem 3.6, so all derivatives at x_* that are defined coincide. Smooth normal-form coefficients depend only on a finite jet at the bifurcation point; hence the center-manifold invariance equations and reduced normal forms are identical. $\square$

Remark 3.8. Corollary 3.7 is a central structural distinction in the present problem. Clipping cannot modify the birth mechanism of a sufficiently small smooth period-two orbit. It can affect that orbit only after its amplitude has grown sufficiently for one of its points to reach Σ_c . The latter is a finite-amplitude event and will be analyzed in Sections 5 and 6.

3.5 Exact local stability threshold

We next determine the stability of x_* . The threshold $\eta_f = 2/\lambda_*$ below is the classical maximal step size of (unclipped) gradient descent [2, 3]; Theorem 3.9 shows that it is preserved exactly by clipping.

Theorem 3.9 (Clipping-independent local stability threshold). *Assume (H1)–(H3) and let*

$$\eta_f = \frac{2}{\lambda_*}.$$

Then, for every fixed $c > 0$, the following statements hold.

(i) If

$$0 < \eta < \eta_f, \quad (3.12)$$

then x_* is a locally asymptotically stable fixed point of $F_{\eta,c}$.

(ii) If

$$\eta > \eta_f, \quad (3.13)$$

then x_* is unstable for $F_{\eta,c}$.

(iii) At

$$\eta = \eta_f, \quad (3.14)$$

the fixed point is nonhyperbolic. Its unique multiplier on the unit circle is the simple multiplier -1 associated with v_* .

Consequently, the first local stability threshold is exactly

$$\eta_f = \frac{2}{\lambda_{\max}(H_*)} \quad (3.15)$$

and is independent of the clipping threshold c .

Proof. By Corollary 3.7,

$$DF_{\eta,c}(x_*) = I - \eta H_*.$$

Since H_* is symmetric positive definite, the multipliers are

$$\rho_i(\eta) = 1 - \eta\lambda_i, \quad i = 1, \dots, d.$$

Suppose first that

$$0 < \eta < \frac{2}{\lambda_*}.$$

Since $0 < \lambda_i \leq \lambda_*$,

$$0 < \eta\lambda_i < 2,$$

and hence

$$-1 < 1 - \eta\lambda_i < 1 \quad \text{for every } i.$$

Thus

$$\max_i |\rho_i(\eta)| < 1.$$

The fixed point is therefore locally asymptotically stable by the standard linearization theorem for discrete dynamical systems.

Now suppose that

$$\eta > \frac{2}{\lambda_*}.$$

Then

$$\rho_d(\eta) = 1 - \eta\lambda_* < -1,$$

so

$$|\rho_d(\eta)| > 1.$$

Hence x_* is unstable.

Finally, if

$$\eta = \eta_f = \frac{2}{\lambda_*},$$

then

$$\rho_d(\eta_f) = -1.$$

By **(H3)**, for every $i < d$,

$$0 < \lambda_i < \lambda_*,$$

and therefore

$$-1 < 1 - \frac{2\lambda_i}{\lambda_*} < 1.$$

Thus -1 is the unique multiplier on the unit circle, and it is simple. $\square$

Remark 3.10. Theorem 3.9 deliberately makes no stability claim at $\eta = \eta_f$. Linearization is inconclusive when the critical multiplier has modulus one. The nonlinear behavior at this parameter is governed by the flip coefficient ℓ_* and will be analyzed in Section 4.

3.6 Local invariant structures

The local identity also transfers invariant structures from the smooth gradient map to the clipped map. For a map T and a radius $r > 0$, define the local stable set relative to $B_r(x_*)$ by

$$W_r^s(x_*; T) := \{x \in B_r(x_*) : T^n(x) \in B_r(x_*) \ \forall n \geq 0, \quad T^n(x) \rightarrow x_*\}. \quad (3.16)$$

Proposition 3.11 (Equality of local stable dynamics). *Let $c > 0$, and choose $0 < r \leq r_c$. Then, for every $\eta > 0$,*

$$W_r^s(x_*; F_{\eta,c}) = W_r^s(x_*; G_\eta). \quad (3.17)$$

Proof. The two maps coincide on $B_r(x_*)$ by Theorem 3.6, so the defining conditions in (3.16) are equivalent for the two maps. $\square$

When x_* is hyperbolic, this equality gives equality of the corresponding stable-manifold germs. If, in addition,

$$\det(I - \eta H_*) \neq 0, \quad (3.18)$$

then both maps are local diffeomorphisms near x_* , and the same argument gives equality of their local unstable-manifold germs.

At the critical parameter $\eta = \eta_f$, the fixed point is nonhyperbolic. Nevertheless, local identity still implies equality of the corresponding center dynamics.

Proposition 3.12 (Center-manifold invariance at the flip threshold). *Assume (H1)–(H3) and set $\eta = \eta_f$. For every fixed $c > 0$, a sufficiently small local center manifold of G_{η_f} through x_* is also a local center manifold of $F_{\eta_f,c}$.*

Moreover, the reduced maps on this common center manifold are identical.

Proof. By Theorem 3.6, the maps $F_{\eta_f,c}$ and G_{η_f} coincide in a neighborhood of x_* . Choose a local center manifold of G_{η_f} contained in that neighborhood. Its local invariance equation is therefore also satisfied by $F_{\eta_f,c}$. Since the restrictions of the two maps to the manifold coincide pointwise, the corresponding reduced one-dimensional dynamics are identical. $\square$

Remark 3.13 (Nonuniqueness of center manifolds). Center manifolds need not be unique as sets; the content of Proposition 3.12 is only that the two maps admit one common choice with identical reduced dynamics.

3.7 Consequences for the smooth-to-switching mechanism

Corollary 3.14 (Local bifurcation structure is clipping-independent). *Under (H1)–(H4), for every fixed $c > 0$:*

- (i) *the fixed point x_* is unchanged by clipping;*
- (ii) *its first local stability threshold is*

$$\eta_f = \frac{2}{\lambda_*};$$
- (iii) *the critical eigendirection is v_* ;*
- (iv) *the critical multiplier crosses -1 transversally at η_f ;*
- (v) *the local center-manifold reduction is identical to that of the unclipped gradient map;*
- (vi) *the supercritical flip coefficient ℓ_* is independent of c .*

Proof. Items (i)–(vi) follow respectively from Proposition 3.1, Theorem 3.9, (2.11), (2.30), Proposition 3.12, and Corollary 3.7. $\square$

The key implication is that c does not enter the local birth mechanism of the period-doubled branch: its first dynamical effect must occur at a distinct finite-amplitude event, when the bifurcating orbit reaches the endogenous switching hypersurface Σ_c .

4 Smooth flip bifurcation and the emerging period-two branch

In this section we study the smooth gradient map

$$G_\eta(x) = x - \eta \nabla V(x)$$

near the critical step size

$$\eta_f = \frac{2}{\lambda_*}.$$

The local mechanism considered here is the classical simple flip bifurcation of a smooth discrete map [5, 6]. For gradient-descent dynamics, Mulayoff and Stich [9] recently derived a multivariate nonlinear stability criterion involving the second-, third-, and fourth-order derivatives of the potential, while Gan [10] formulated edge-of-stability dynamics in a systematic bifurcation-theoretic framework.

Accordingly, the existence of a smooth period-two branch is not the new contribution of the present paper. We reproduce the local reduction because its explicit coefficient enters the switching-separation law derived later; we therefore give a self-contained center-manifold and normal-form derivation in the notation of the present paper.

By the local identity theorem established in Section 3, the resulting local bifurcation structure is inherited exactly by the clipped map $F_{\eta,c}$ as long as the bifurcating orbit remains inside the smooth region Ω_c^- .

Throughout this section we write

$$\mu = \eta - \eta_f. \quad (4.1)$$

Thus the bifurcation point corresponds to $\mu = 0$.

Under **(H2)**–**(H3)**, the multiplier associated with v_* is

$$\rho_*(\mu) = 1 - (\eta_f + \mu)\lambda_* = -1 - \lambda_*\mu, \quad (4.2)$$

whereas all remaining multipliers lie strictly inside the unit disk at $\mu = 0$.

The nonlinear character of the bifurcation is governed by the coefficient

$$\ell_* = \frac{\Gamma_*}{3\lambda_*},$$

defined in (2.24). Assumption **(H4)** gives

$$\ell_* > 0,$$

which corresponds to the supercritical case.

4.1 Parameter-dependent center reduction

Translate the critical point to the origin, $y = x - x_*$, and define

$$\Phi_\mu(y) := G_{\eta_f + \mu}(x_* + y) - x_*. \quad (4.3)$$

Then $\Phi_\mu(0) = 0$ for every sufficiently small μ , and at $\mu = 0$,

$$D\Phi_0(0) = I - \eta_f H_*, \quad (4.4)$$

whose unique critical multiplier is -1 with eigenvector v_* ; the complementary spectrum lies strictly inside the unit disk.

To incorporate the bifurcation parameter, consider the extended map

$$(y, \mu) \mapsto (\Phi_\mu(y), \mu). \quad (4.5)$$

Its two-dimensional local center manifold is associated with the critical state direction v_* and the neutral parameter direction, and after restricting to a sufficiently small neighborhood can be represented as

$$y = zv_* + h(z, \mu), \quad (4.6)$$

with

$$h(z, \mu) \in v_*^\perp, \quad h(0, \mu) = 0, \quad D_z h(0, 0) = 0, \quad h(z, \mu) = O(z^2 + |\mu||z|). \quad (4.7)$$

The dynamics on the center manifold are described by the scalar map

$$z_{n+1} = f_\mu(z_n). \quad (4.8)$$

4.2 Flip normal form

We next record the local normal form required for the asymptotic analysis.

Proposition 4.1 (Reduced flip normal form). *Assume **(H1)**–**(H4)**. There exist local state and parameter coordinates, preserving $\mu = 0$ and $z = 0$, in which the reduced map (4.8) takes the form*

$$f_\mu(z) = -(1 + \lambda_*\mu)z + \ell_*z^3 + R(z, \mu), \quad (4.9)$$

where

$$R(z, \mu) = O(|z|^4 + |\mu||z|^2 + \mu^2|z|) \quad (4.10)$$

as $(z, \mu) \rightarrow (0, 0)$.

The cubic coefficient is

$$\ell_* = \frac{\eta_f}{6} (3T_*[v_*, v_*, H_*^{-1}g_*] - Q_*[v_*, v_*, v_*, v_*]), \quad (4.11)$$

or equivalently

$$\ell_* = \frac{\Gamma_*}{3\lambda_*}. \quad (4.12)$$

Proof. The critical multiplier -1 is simple and crosses the unit circle transversally (Section 2.5), so the standard parameter-dependent center-manifold and flip-normal-form reductions apply; see [37, 38, 5]. The linear parameter coefficient is $\rho_*(\mu) = -1 - \lambda_*\mu$ by (4.2), and the quadratic term is eliminated by the standard near-identity transformation for a simple flip; the remaining cubic coefficient is the first flip coefficient.

For the gradient map

$$G_\eta(x) = x - \eta \nabla V(x),$$

one has

$$D^2G_{\eta_f}(x_*)[u, w] = -\eta_f T_*[u, w, \cdot]^\sharp, \quad (4.13)$$

$$D^3G_{\eta_f}(x_*)[u, w, q] = -\eta_f Q_*[u, w, q, \cdot]^\sharp. \quad (4.14)$$

The standard flip-coefficient formula therefore gives

$$\ell_* = \frac{\eta_f}{2} T_*[v_*, v_*, H_*^{-1}g_*] - \frac{\eta_f}{6} Q_*[v_*, v_*, v_*, v_*].$$

Using

$$\eta_f = \frac{2}{\lambda_*}$$

and the definition of Γ_* in (2.22) yields

$$\ell_* = \frac{\Gamma_*}{3\lambda_*}.$$

The remainder estimate (4.10) follows from the regularity in **(H1)** and the parameter-dependent near-identity normal-form transformation. $\square$

Remark 4.2. The sign convention in (4.9) is fixed throughout the paper:

$$\ell_* > 0$$

corresponds to a supercritical flip bifurcation, with a stable period-two branch emerging for $\mu > 0$. This convention is consistent with the multivariate gradient-descent flip criterion encoded by **(H4)**.

4.3 The second iterate

The bifurcating period-two orbit is most conveniently obtained from the fixed points of the second iterate.

Lemma 4.3 (Expansion of the second iterate). *Under the hypotheses of Proposition 4.1,*

$$f_\mu^2(z) - z = 2\lambda_*\mu z - 2\ell_*z^3 + \mathcal{R}_2(z, \mu), \quad (4.15)$$

where

$$\mathcal{R}_2(z, \mu) = O(|z|^4 + |\mu||z|^2 + \mu^2|z|). \quad (4.16)$$

Proof. Write

$$a_\mu := 1 + \lambda_* \mu.$$

Then

$$f_\mu(z) = -a_\mu z + \ell_* z^3 + R(z, \mu).$$

Ignoring terms of order contained in (4.16), one obtains

$$\begin{aligned} f_\mu^2(z) &= -a_\mu (-a_\mu z + \ell_* z^3) + \ell_* (-a_\mu z)^3 + \mathcal{R}_2(z, \mu) \\ &= a_\mu^2 z - \ell_* a_\mu (1 + a_\mu^2) z^3 + \mathcal{R}_2(z, \mu). \end{aligned} \quad (4.17)$$

Since

$$a_\mu^2 = 1 + 2\lambda_* \mu + O(\mu^2)$$

and

$$a_\mu(1 + a_\mu^2) = 2 + O(\mu),$$

we obtain

$$f_\mu^2(z) - z = 2\lambda_* \mu z - 2\ell_* z^3 + \mathcal{R}_2(z, \mu).$$

The stated remainder estimate follows by substituting (4.10) into the composition. $\square$

Equation (4.15) exhibits explicitly the balance responsible for the square-root amplitude law:

$$\lambda_* \mu \sim \ell_* z^2.$$

4.4 Existence and amplitude of the period-two branch

We now obtain the local period-two branch together with its leading amplitude.

Theorem 4.4 (Supercritical period-two branch). *Assume (H1)–(H4). There exists $\mu_0 > 0$ such that for every*

$$0 < \mu < \mu_0,$$

the smooth gradient map $G_{\eta_f + \mu}$ possesses a unique local period-two cycle

$$\mathcal{O}_2(\mu) = \{x_-(\mu), x_+(\mu)\} \quad (4.18)$$

in a sufficiently small neighborhood of x_ .*

The two points satisfy

$$x_\pm(\mu) = x_* \pm A_* \mu^{1/2} v_* + O(\mu), \quad \mu \rightarrow 0^+, \quad (4.19)$$

where

$$A_* = \sqrt{\frac{\lambda_*}{\ell_*}} \quad (4.20)$$

and therefore

$$A_*^2 = \frac{3\lambda_*^2}{\Gamma_*}. \quad (4.21)$$

The labeling may be chosen so that

$$G_{\eta_f + \mu}(x_+(\mu)) = x_-(\mu), \quad G_{\eta_f + \mu}(x_-(\mu)) = x_+(\mu). \quad (4.22)$$

Proof. We first work in the reduced coordinate z .

Set

$$\mu = \varepsilon^2, \quad \varepsilon > 0,$$

and seek nonzero solutions of

$$f_{\varepsilon^2}^2(z) - z = 0$$

in the form

$$z = \varepsilon w.$$

By Lemma 4.3,

$$\begin{aligned} f_{\varepsilon^2}^2(\varepsilon w) - \varepsilon w &= 2\lambda_* \varepsilon^3 w - 2\ell_* \varepsilon^3 w^3 + O(\varepsilon^4) \\ &= 2\varepsilon^3 [w(\lambda_* - \ell_* w^2) + O(\varepsilon)]. \end{aligned} \quad (4.23)$$

Define the rescaled bifurcation equation

$$\mathcal{F}(w, \varepsilon) := \frac{f_{\varepsilon^2}^2(\varepsilon w) - \varepsilon w}{2\varepsilon^3}, \quad \varepsilon > 0. \quad (4.24)$$

Under **(H1)** the rescaled map extends as a C^3 function to $\varepsilon = 0$: since $V \in C^7$, the gradient map G_η is C^6 , and by the parameter-dependent flip reduction so is the reduced map f_μ ; the numerator in (4.24) is therefore C^6 in (ε, w) and vanishes to third order in ε , so dividing by $2\varepsilon^3$ yields a C^3 function. We set

$$\mathcal{F}(w, 0) = w(\lambda_* - \ell_* w^2). \quad (4.25)$$

Since $\ell_* > 0$, the nonzero zeros of $\mathcal{F}(\cdot, 0)$ are

$$w_\pm^0 = \pm \sqrt{\frac{\lambda_*}{\ell_*}}. \quad (4.26)$$

Furthermore,

$$\begin{aligned} \partial_w \mathcal{F}(w_\pm^0, 0) &= \lambda_* - 3\ell_* (w_\pm^0)^2 \\ &= -2\lambda_* \neq 0. \end{aligned} \quad (4.27)$$

The implicit function theorem therefore yields two branches

$$w_\pm(\varepsilon) = w_\pm^0 + O(\varepsilon).$$

Consequently,

$$\begin{aligned} z_\pm(\mu) &= \varepsilon w_\pm(\varepsilon) \\ &= \pm \sqrt{\frac{\lambda_*}{\ell_*}} \sqrt{\mu} + O(\mu). \end{aligned} \quad (4.28)$$

The fixed point $z = 0$ is locally the only fixed point of f_μ near the origin. Indeed,

$$\partial_z(f_0(z) - z)|_{z=0} = -2 \neq 0,$$

so the implicit function theorem excludes nearby nonzero period-one fixed points. Hence the two nonzero solutions $z_\pm(\mu)$ of

$$f_\mu^2(z) = z$$

have prime period two.

Returning to the original coordinates through (4.6),

$$x_\pm(\mu) = x_* + z_\pm(\mu)v_* + h(z_\pm(\mu), \mu).$$

Since

$$h(z, \mu) = O(z^2 + |\mu||z|)$$

and

$$z_\pm(\mu) = O(\sqrt{\mu}),$$

we have

$$h(z_\pm(\mu), \mu) = O(\mu).$$

Therefore

$$x_\pm(\mu) = x_* \pm \sqrt{\frac{\lambda_*}{\ell_*}} \sqrt{\mu} v_* + O(\mu),$$

which proves (4.19).

Finally, the two nonzero points form a single two-cycle, so after possibly interchanging the labels we obtain (4.22).

The uniqueness of the sufficiently small period-two cycle follows from the local center-manifold reduction and the uniqueness of the two nonzero solution branches of the reduced second-iterate equation. $\square$

4.5 Stability of the emerging cycle

We next verify the local stability of the branch.

Theorem 4.5 (Stability of the supercritical period-two cycle). *Under (H1)–(H4), the period-two cycle $\mathcal{O}_2(\mu)$ given by Theorem 4.4 is locally asymptotically stable for all sufficiently small $\mu > 0$.*

Along the critical center direction, the multiplier of the two-step map satisfies

$$\rho_{\text{cyc}}^c(\mu) = 1 - 4\lambda_*\mu + O(\mu^{3/2}), \quad \mu \rightarrow 0^+. \quad (4.29)$$

Proof. Differentiate the expansion (4.15):

$$(f_\mu^2)'(z) = 1 + 2\lambda_*\mu - 6\ell_*z^2 + O(|z|^3 + |\mu||z| + \mu^2). \quad (4.30)$$

At either point $z = z_\pm(\mu)$, Theorem 4.4 gives

$$z_\pm(\mu)^2 = \frac{\lambda_*}{\ell_*}\mu + O(\mu^{3/2}).$$

Hence

$$\begin{aligned} (f_\mu^2)'(z_\pm(\mu)) &= 1 + 2\lambda_*\mu - 6\lambda_*\mu + O(\mu^{3/2}) \\ &= 1 - 4\lambda_*\mu + O(\mu^{3/2}). \end{aligned} \quad (4.31)$$

Thus the center multiplier lies strictly inside the unit disk for all sufficiently small $\mu > 0$.

At $\mu = 0$, every transverse multiplier of the two-step map equals

$$\rho_i(\eta_f)^2, \quad i = 1, \dots, d-1,$$

and by (2.28),

$$|\rho_i(\eta_f)| < 1.$$

Therefore

$$\rho_i(\eta_f)^2 < 1.$$

By continuity, all transverse multipliers remain strictly inside the unit disk along the sufficiently small period-two branch.

The reduction principle then implies local asymptotic stability of the full period-two cycle. $\square$

Remark 4.6. Equation (4.29) shows that the newly created cycle is only weakly attracting immediately after the flip:

$$\rho_{\text{cyc}}^c(\mu) \rightarrow 1^- \quad \text{as } \mu \rightarrow 0^+.$$

This weak attraction is relevant later when the branch approaches the switching hypersurface, because the switching interaction occurs on a parameter scale that also tends to zero with c .

4.6 Transfer to the clipped map before switching

We finally transfer the preceding smooth bifurcation result to the clipped map.

Corollary 4.7 (Pre-contact period-two cycle for the clipped map). *Assume (H1)–(H5) and fix $c > 0$. Then there exists*

$$\mu_c > 0$$

such that for every

$$0 < \mu < \mu_c,$$

the clipped map

$$F_{\eta_f + \mu, c}$$

possesses the same locally asymptotically stable period-two cycle $\mathcal{O}_2(\mu)$ as the smooth gradient map $G_{\eta_f + \mu}$.

Moreover,

$$x_\pm(\mu) \in \Omega_c^-, \quad 0 < \mu < \mu_c. \quad (4.32)$$

Proof. For the fixed value $c > 0$, let $r_c > 0$ be given by Theorem 3.6. By Theorem 4.4,

$$x_{\pm}(\mu) \rightarrow x_* \quad \text{as } \mu \rightarrow 0^+.$$

Hence there exists $\mu_c > 0$ such that

$$x_{\pm}(\mu) \in B_{r_c}(x_*) \subset \Omega_c^-$$

for every $0 < \mu < \mu_c$.

On this ball,

$$F_{\eta_f + \mu, c} = G_{\eta_f + \mu}$$

by Theorem 3.6. Therefore the two maps have exactly the same period-two cycle and the same local derivatives along that cycle. Its local asymptotic stability follows from Theorem 4.5. $\square$

Remark 4.8 (The first possible effect of clipping). Corollary 4.7 shows that the period-two branch is born entirely inside the smooth region. Therefore clipping does not influence either its creation or its initial stability.

The first possible effect of clipping occurs only when one of the points $x_{\pm}(\mu)$ reaches the endogenous switching hypersurface

$$\Sigma_c = \{x : \|\nabla V(x)\|_{\diamond} = c\}.$$

Determining the parameter value of this first contact is the central problem of Sections 5 and 6.

5 Switching geometry along the period-two branch

We now study how the smooth period-two branch constructed in Section 4 approaches the endogenous switching hypersurface

$$\Sigma_c = \{x \in \mathcal{U} : \|\nabla V(x)\|_{\diamond} = c\}.$$

A key structural feature of gradient maps appears at this point. For an arbitrary period-two cycle of gradient descent, the gradients at its two points are exactly opposite. Consequently, their gradient norms are identical for every norm $\|\cdot\|_{\diamond}$.

This identity has two important consequences.

First, the two points of the smooth period-two orbit necessarily reach the clipping boundary simultaneously. Thus, unlike a generic piecewise-smooth map, the present gradient-generated switching problem does not exhibit a splitting of the two first-contact parameters.

Second, the common gradient magnitude along the bifurcating cycle has a particularly simple expansion in the square-root parameter $\varepsilon = \sqrt{\eta - \eta_f}$. This structure will yield the quadratic separation law in Section 6.

5.1 Exact gradient balance on a period-two cycle

We begin with an exact identity that does not rely on the local bifurcation expansion. An equivalent balance relation appears in the recent characterization of gradient-descent period-two orbits by Litman [11]. Since this identity is the starting point of the switching analysis below, we include its elementary proof for completeness.

Proposition 5.1 (Exact gradient balance). *Let $\eta > 0$, and suppose that $\{x_-, x_+\}$ is a period-two cycle of the smooth gradient map*

$$G_{\eta}(x) = x - \eta \nabla V(x),$$

that is,

$$G_{\eta}(x_+) = x_-, \quad G_{\eta}(x_-) = x_+, \quad x_+ \neq x_-. \quad (5.1)$$

Then

$$\nabla V(x_+) = \frac{x_+ - x_-}{\eta}, \quad \nabla V(x_-) = -\frac{x_+ - x_-}{\eta}. \quad (5.2)$$

In particular,

$$\nabla V(x_+) = -\nabla V(x_-). \quad (5.3)$$

Consequently, for every norm $\|\cdot\|_{\diamond}$,

$$\|\nabla V(x_+)\|_{\diamond} = \|\nabla V(x_-)\|_{\diamond}. \quad (5.4)$$

Proof. The first identity in (5.1) gives

$$x_- = x_+ - \eta \nabla V(x_+),$$

and hence

$$\nabla V(x_+) = \frac{x_+ - x_-}{\eta}.$$

Similarly,

$$x_+ = x_- - \eta \nabla V(x_-)$$

implies

$$\nabla V(x_-) = \frac{x_- - x_+}{\eta} = -\frac{x_+ - x_-}{\eta}.$$

This proves (5.2) and (5.3).

Since every norm is even,

$$\| -g \|_{\diamond} = \| g \|_{\diamond},$$

and therefore

$$\| \nabla V(x_-) \|_{\diamond} = \| -\nabla V(x_+) \|_{\diamond} = \| \nabla V(x_+) \|_{\diamond}.$$

□

Remark 5.2 (Gradient structure versus a generic switching map). Equation (5.4) is a special consequence of the gradient-descent structure. In a generic piecewise-smooth map, the two points of a period-two orbit need not have equal switching functions and may therefore encounter a switching boundary at different parameter values.

For radial gradient clipping, by contrast, the switching function depends only on

$$\| \nabla V(x) \|_{\diamond},$$

and Proposition 5.1 forces the two switching values to be identical along every smooth period-two cycle.

5.2 Midpoint and half-difference coordinates

For the bifurcating period-two cycle

$$\mathcal{O}_2(\mu) = \{x_-(\mu), x_+(\mu)\}, \quad \mu = \eta - \eta_f > 0,$$

define its midpoint and half-difference by

$$m(\mu) := \frac{x_+(\mu) + x_-(\mu)}{2} - x_*, \tag{5.5}$$

$$d(\mu) := \frac{x_+(\mu) - x_-(\mu)}{2}. \tag{5.6}$$

Thus

$$x_{\pm}(\mu) = x_* + m(\mu) \pm d(\mu). \tag{5.7}$$

Theorem 4.4 implies

$$m(\mu) = O(\mu), \quad d(\mu) = A_* \sqrt{\mu} v_* + O(\mu). \tag{5.8}$$

The period-two equations yield considerably more structure.

Lemma 5.3 (Midpoint–difference equations). *For every smooth period-two cycle of G_{η} written in the form (5.7),*

$$\nabla V(x_* + m + d) + \nabla V(x_* + m - d) = 0, \tag{5.9}$$

$$\nabla V(x_* + m + d) - \nabla V(x_* + m - d) = \frac{4}{\eta} d. \tag{5.10}$$

Proof. By Proposition 5.1,

$$\nabla V(x_* + m + d) = \frac{2}{\eta} d$$

and

$$\nabla V(x_* + m - d) = -\frac{2}{\eta} d.$$

Adding and subtracting these identities gives (5.9) and (5.10). □

5.3 Square-root parametrization and parity

By the parameter-dependent flip theorem [5, 6] and **(H1)**, the local period-two branch admits a C^4 signed parametrization in the variable ε introduced below; in particular the midpoint and half-difference functions are C^4 , and the parity relations derived next are justified to the orders used in Proposition 5.4.

Introduce the signed square-root parameter

$$\varepsilon^2 = \mu, \quad \eta = \eta_f + \varepsilon^2. \quad (5.11)$$

The two-cycle equations are invariant under the transformation

$$(d, \varepsilon) \mapsto (-d, -\varepsilon),$$

which merely exchanges the two points of the cycle.

The local uniqueness of the bifurcating period-two branch therefore allows the branch to be parametrized so that

$$m(-\varepsilon) = m(\varepsilon), \quad d(-\varepsilon) = -d(\varepsilon). \quad (5.12)$$

Hence the midpoint is even and the half-difference is odd in ε .

Proposition 5.4 (Refined two-cycle expansion). *Assume **(H1)**–**(H4)**. The period-two branch can be parametrized by the signed variable ε , with $\mu = \varepsilon^2$, such that*

$$m(\varepsilon) = \varepsilon^2 m_2 + O(\varepsilon^4), \quad (5.13)$$

$$d(\varepsilon) = A_* \varepsilon v_* + O(\varepsilon^3), \quad (5.14)$$

where

$$m_2 = -\frac{A_*^2}{2} H_*^{-1} g_* = -\frac{A_*^2}{2} h_*. \quad (5.15)$$

Consequently,

$$x_{\pm}(\mu) = x_* - \frac{A_*^2}{2} \mu h_* \pm A_* \sqrt{\mu} v_* + O(\mu^{3/2}). \quad (5.16)$$

Proof. The parity relations (5.12) imply that the local expansions have the form

$$m(\varepsilon) = \varepsilon^2 m_2 + O(\varepsilon^4)$$

and

$$d(\varepsilon) = \varepsilon d_1 + O(\varepsilon^3).$$

By Theorem 4.4,

$$d_1 = A_* v_*.$$

It remains to determine m_2 . Taylor expansion of the gradient about x_* gives

$$\nabla V(x_* + y) = H_* y + \frac{1}{2} T_*[y, y, \cdot]^\sharp + O(\|y\|_2^3). \quad (5.17)$$

Apply (5.17) to $y = m + d$ and $y = m - d$ and add the resulting expressions. Using (5.9), we obtain

$$2H_* m + T_*[d, d, \cdot]^\sharp + O(\varepsilon^4) = 0. \quad (5.18)$$

Here we used

$$m = O(\varepsilon^2), \quad d = O(\varepsilon),$$

and the cancellation of all terms odd in d .

Substituting

$$m = \varepsilon^2 m_2 + O(\varepsilon^4)$$

and

$$d = A_* \varepsilon v_* + O(\varepsilon^3)$$

into (5.18) gives

$$2H_* m_2 + A_*^2 T_*[v_*, v_*, \cdot]^\sharp = 0.$$

Since H_* is invertible,

$$m_2 = -\frac{A_*^2}{2} H_*^{-1} T_*[v_*, v_*, \cdot]^\sharp = -\frac{A_*^2}{2} h_*.$$

This proves (5.15) and hence (5.16). $\square$

Remark 5.5 (Nonlinear displacement of the cycle midpoint). Although the leading oscillation is aligned with the maximal Hessian eigendirection v_* , the midpoint of the bifurcating two-cycle generally does not remain at x_* . Instead,

$$\frac{x_+(\mu) + x_-(\mu)}{2} = x_* - \frac{A_*^2}{2}\mu h_* + O(\mu^2).$$

Thus the third derivative of V produces an $O(\mu)$ drift of the cycle center through the vector

$$h_* = H_*^{-1}T_*[v_*, v_*, \cdot]^\sharp.$$

5.4 A consistency derivation of the flip amplitude

The midpoint–difference equations also provide a direct, coordinate-free derivation of the leading oscillation amplitude.

Proposition 5.6 (Amplitude identity). *Under (H1)–(H4), the leading amplitude satisfies*

$$A_*^2\Gamma_* = 3\lambda_*^2. \quad (5.19)$$

Equivalently,

$$A_*^2 = \frac{3\lambda_*^2}{\Gamma_*}, \quad (5.20)$$

in agreement with (4.21).

Proof. Expand the difference equation (5.10) to cubic order.

Using

$$m = \varepsilon^2 m_2 + O(\varepsilon^4), \quad d = A_* \varepsilon v_* + O(\varepsilon^3),$$

Taylor expansion gives

$$\begin{aligned} \nabla V(x_* + m + d) - \nabla V(x_* + m - d) \\ = 2H_* d + 2T_*[m, d, \cdot]^\sharp + \frac{1}{3}Q_*[d, d, d, \cdot]^\sharp + O(\varepsilon^4). \end{aligned} \quad (5.21)$$

On the other hand,

$$\frac{4}{\eta} = \frac{4}{\eta_f + \varepsilon^2} = \frac{4}{\eta_f} - \frac{4}{\eta_f^2}\varepsilon^2 + O(\varepsilon^4).$$

Take the Euclidean inner product of (5.10) with v_* and compare the coefficients of ε^3 .

The unknown $O(\varepsilon^3)$ correction in d cancels because

$$\frac{4}{\eta_f} = 2\lambda_*.$$

Thus,

$$2A_*T_*[v_*, v_*, m_2] + \frac{A_*^3}{3}Q_*[v_*, v_*, v_*, v_*] = -\frac{4A_*}{\eta_f^2}. \quad (5.22)$$

Since

$$m_2 = -\frac{A_*^2}{2}h_*$$

and

$$\eta_f = \frac{2}{\lambda_*},$$

equation (5.22) becomes

$$-A_*^3T_*[v_*, v_*, h_*] + \frac{A_*^3}{3}Q_*[v_*^4] = -A_*\lambda_*^2.$$

Dividing by $A_* > 0$ and rearranging gives

$$A_*^2(3T_*[v_*, v_*, h_*] - Q_*[v_*^4]) = 3\lambda_*^2.$$

By definition of Γ_* ,

$$A_*^2\Gamma_* = 3\lambda_*^2.$$

□

Remark 5.7. Proposition 5.6 provides an independent check of the center-manifold normal-form calculation in Section 4. In particular, the same high-order quantity Γ_* governs both the supercriticality criterion and the leading amplitude of the emerging period-two cycle.

5.5 Gradient exposure along the bifurcating cycle

Define the common gradient exposure of the smooth period-two cycle by

$$\mathcal{E}(\mu) := \|\nabla V(x_+(\mu))\|_\diamond = \|\nabla V(x_-(\mu))\|_\diamond. \quad (5.23)$$

The equality is exact by Proposition 5.1.

Using the half-difference coordinate, one obtains the exact formula

$$\mathcal{E}(\mu) = \frac{2}{\eta_f + \mu} \|d(\mu)\|_\diamond. \quad (5.24)$$

This identity is considerably stronger than a direct Taylor expansion of ∇V at the two cycle points.

Theorem 5.8 (Asymptotic switching geometry of the two-cycle). *Assume (H1)–(H5). As $\mu \rightarrow 0^+$,*

$$\mathcal{E}(\mu) = B_* \mu^{1/2} + O(\mu^{3/2}), \quad (5.25)$$

where

$$B_* = \lambda_* A_* \|v_*\|_\diamond. \quad (5.26)$$

Equivalently,

$$B_*^2 = \frac{3\lambda_*^4 \|v_*\|_\diamond^2}{\Gamma_*}. \quad (5.27)$$

In particular, there is no term of order μ in (5.25).

Proof. Set

$$\varepsilon = \sqrt{\mu}.$$

By Proposition 5.4,

$$d(\varepsilon) = A_* \varepsilon v_* + O(\varepsilon^3).$$

Using positive homogeneity of the norm,

$$\begin{aligned} \|d(\varepsilon)\|_\diamond &= \varepsilon \|A_* v_* + O(\varepsilon^2)\|_\diamond \\ &= A_* \|v_*\|_\diamond \varepsilon + O(\varepsilon^3). \end{aligned} \quad (5.28)$$

Furthermore,

$$\frac{2}{\eta_f + \varepsilon^2} = \frac{2}{\eta_f} + O(\varepsilon^2) = \lambda_* + O(\varepsilon^2). \quad (5.29)$$

Substituting (5.28) and (5.29) into (5.24) yields

$$\mathcal{E}(\mu) = \lambda_* A_* \|v_*\|_\diamond \varepsilon + O(\varepsilon^3).$$

Since $\varepsilon = \sqrt{\mu}$,

$$\mathcal{E}(\mu) = B_* \mu^{1/2} + O(\mu^{3/2}),$$

where

$$B_* = \lambda_* A_* \|v_*\|_\diamond.$$

Finally, using

$$A_*^2 = \frac{3\lambda_*^2}{\Gamma_*}$$

gives

$$B_*^2 = \lambda_*^2 A_*^2 \|v_*\|_\diamond^2 = \frac{3\lambda_*^4 \|v_*\|_\diamond^2}{\Gamma_*}.$$

□

Remark 5.9 (Absence of a quadratic term in the exposure). A naive substitution of

$$x_{\pm}(\mu) = x_* \pm A_* \sqrt{\mu} v_* + O(\mu)$$

into the gradient norm might suggest an expansion of the form

$$B_1 \sqrt{\mu} + B_2 \mu + \dots$$

The exact two-cycle identity (5.24) shows that this is not the case.

The half-difference d is odd in the signed square-root parameter ε , while $\eta = \eta_f + \varepsilon^2$ is even. Consequently, the common exposure is an odd function of ε to leading orders:

$$\mathcal{E}(\varepsilon^2) = B_* \varepsilon + O(\varepsilon^3).$$

This structural cancellation will improve the remainder in the switching-contact separation law from the naively expected $O(c^3)$ to $O(c^4)$.

5.6 Simultaneous first contact

We now record the most important geometric consequence of the exact gradient balance.

Theorem 5.10 (Simultaneous switching contact). *Let $\mathcal{O}_2(\mu) = \{x_-(\mu), x_+(\mu)\}$ be the smooth period-two branch from Theorem 4.4. For every sufficiently small $\mu > 0$,*

$$x_+(\mu) \in \Sigma_c \iff x_-(\mu) \in \Sigma_c. \quad (5.30)$$

Likewise,

$$x_+(\mu) \in \Omega_c^- \iff x_-(\mu) \in \Omega_c^-, \quad (5.31)$$

$$x_+(\mu) \in \Omega_c^+ \iff x_-(\mu) \in \Omega_c^+. \quad (5.32)$$

Hence the two points of a smooth gradient-descent period-two cycle cannot cross a radial clipping boundary sequentially.

Proof. By Proposition 5.1,

$$\|\nabla V(x_+(\mu))\|_{\diamond} = \|\nabla V(x_-(\mu))\|_{\diamond}.$$

Each of the three statements follows immediately by comparing this common value with the clipping threshold c . $\square$

Corollary 5.11 (No pre-contact splitting of the two-cycle). *There do not exist two distinct first-contact parameters*

$$\eta_{sc}^+(c) \neq \eta_{sc}^-(c)$$

for the two points of the smooth period-two branch. Whenever the first switching contact exists, it is a simultaneous two-point contact.

Proof. If one point lies on Σ_c , then the other lies on Σ_c by Theorem 5.10. $\square$

5.7 Impossibility of a mixed smooth–clipped two-cycle

The simultaneous-contact structure is not restricted to the smooth branch before contact. Radial clipping itself rules out a genuine period-two cycle having one point in the smooth region and the other in the clipped region.

Proposition 5.12 (No mixed-itinerary period-two cycle). *Let*

$$\{y_-, y_+\}$$

be a period-two cycle of the clipped map

$$F_{\eta,c}(x) = x - \eta \mathcal{C}_c^{\circ}(\nabla V(x)).$$

Then it is impossible to have

$$y_- \in \Omega_c^-, \quad y_+ \in \Omega_c^+, \quad (5.33)$$

or vice versa.

Therefore every period-two cycle of $F_{\eta,c}$ is either entirely smooth, entirely clipped, or lies on the switching boundary.

Proof. The period-two relations give

$$\mathcal{C}_c^\diamond(\nabla V(y_+)) = \frac{y_+ - y_-}{\eta}, \quad (5.34)$$

$$\mathcal{C}_c^\diamond(\nabla V(y_-)) = -\frac{y_+ - y_-}{\eta}. \quad (5.35)$$

Hence

$$\|\mathcal{C}_c^\diamond(\nabla V(y_+))\|_\diamond = \|\mathcal{C}_c^\diamond(\nabla V(y_-))\|_\diamond. \quad (5.36)$$

For radial clipping,

$$\|\mathcal{C}_c^\diamond(g)\|_\diamond = \min\{\|g\|_\diamond, c\}. \quad (5.37)$$

Suppose, for contradiction, that

$$y_- \in \Omega_c^-, \quad y_+ \in \Omega_c^+.$$

Then

$$\|\mathcal{C}_c^\diamond(\nabla V(y_-))\|_\diamond = \|\nabla V(y_-)\|_\diamond < c,$$

whereas

$$\|\mathcal{C}_c^\diamond(\nabla V(y_+))\|_\diamond = c.$$

This contradicts (5.36). The opposite mixed itinerary is ruled out identically. $\square$

Remark 5.13 (Structural consequence for the post-contact problem). Proposition 5.12 rules out the mixed smooth-clipped period-two continuation that one might expect from a generic switching system.

Thus the post-contact problem for radial gradient clipping is more constrained: if a period-two branch persists beyond the switching contact, both points must enter the clipped regime together.

The corresponding post-contact continuation problem will therefore be formulated as a simultaneous two-point switching bifurcation, rather than as a mixed-itinerary continuation.

5.8 Summary of the switching geometry

The preceding analysis establishes three structural facts that will be used in the next section.

First, the period-two gradients are exactly opposite:

$$\nabla V(x_+(\mu)) = -\nabla V(x_-(\mu)). \quad (5.38)$$

Second, their common clipping exposure satisfies

$$\mathcal{E}(\mu) = \lambda_* A_* \|v_*\|_\diamond \sqrt{\mu} + O(\mu^{3/2}). \quad (5.39)$$

Third, the first switching interaction is necessarily simultaneous at both points of the cycle.

Therefore the first-contact condition reduces to the single scalar equation

$$\mathcal{E}(\mu_{\text{sc}}(c)) = c. \quad (5.40)$$

Section 6 will invert (5.40) and derive the resulting separation law between the smooth flip threshold η_f and the first switching-contact parameter.

6 The flip-to-switching separation law

We now determine the parameter value at which the smooth period-two branch created at the flip bifurcation first reaches the gradient-clipping boundary.

Recall that

$$\mu = \eta - \eta_f, \quad \eta_f = \frac{2}{\lambda_*},$$

and let

$$\mathcal{O}_2(\mu) = \{x_-(\mu), x_+(\mu)\}$$

denote the smooth period-two branch constructed in Theorem 4.4.

By Proposition 5.1, the two points of the cycle have exactly opposite gradients:

$$\nabla V(x_+(\mu)) = -\nabla V(x_-(\mu)).$$

Hence they have a common clipping exposure

$$\mathcal{E}(\mu) := \|\nabla V(x_+(\mu))\|_\diamond = \|\nabla V(x_-(\mu))\|_\diamond. \quad (6.1)$$

The first switching contact therefore occurs when the single scalar condition

$$\mathcal{E}(\mu) = c \quad (6.2)$$

is satisfied.

The main purpose of this section is to solve (6.2) asymptotically as $c \rightarrow 0^+$.

Quadratic tangency between period-doubling and border-collision loci is known for generic piecewise-smooth maps [33]. Accordingly, the exponent two obtained below is not claimed as a generic new border-collision phenomenon.

The contribution here is the explicit gradient-induced coefficient, the simultaneous nature of the two-point contact, and the improved even-power remainder resulting from the exact gradient-balance structure.

6.1 Factorization of the gradient exposure

The square-root law in Theorem 5.8 can be strengthened into a factorized form that is particularly convenient for inversion.

Lemma 6.1 (Exposure factorization). *Assume (H1)–(H5). There exist $\mu_0 > 0$ and a positive C^1 function*

$$\beta : [0, \mu_0) \rightarrow (0, \infty)$$

such that

$$E(\mu) = \sqrt{\mu} \beta(\mu), \quad 0 \leq \mu < \mu_0,$$

with

$$\beta(0) = B_* = \lambda_* A_* \|v_*\|_\diamond.$$

Consequently,

$$B_*^2 = \frac{3\lambda_*^4 \|v_*\|_\diamond^2}{\Gamma_*}.$$

Proof. Introduce the signed square-root parameter

$$\varepsilon^2 = \mu.$$

By Proposition 5.4, the half-difference of the period-two orbit satisfies

$$d(-\varepsilon) = -d(\varepsilon)$$

and

$$d(\varepsilon) = A_* \varepsilon v_* + O(\varepsilon^3).$$

The signed parametrization of Section 5.3 shows that d is a C^3 odd function of ε . Hence it can be written locally as

$$d(\varepsilon) = \varepsilon q(\varepsilon^2), \quad (6.3)$$

where q is a C^1 function and

$$q(0) = A_* v_*. \quad (6.4)$$

By the exact gradient-balance identity (5.24),

$$\mathcal{E}(\mu) = \frac{2}{\eta_f + \mu} \|d(\sqrt{\mu})\|_\diamond.$$

Substituting (6.3) gives

$$\mathcal{E}(\mu) = \sqrt{\mu} \frac{2}{\eta_f + \mu} \|q(\mu)\|_\diamond.$$

Thus define

$$\beta(\mu) := \frac{2}{\eta_f + \mu} \|q(\mu)\|_\diamond. \quad (6.5)$$

Since

$$q(0) = A_* v_* \neq 0$$

and the norm $\|\cdot\|_\diamond$ is smooth away from the origin by (H5), the function β is C^1 after reducing μ_0 if necessary.

Furthermore,

$$\begin{aligned} \beta(0) &= \frac{2}{\eta_f} \|A_* v_*\|_\diamond \\ &= \lambda_* A_* \|v_*\|_\diamond = B_*, \end{aligned} \tag{6.6}$$

because

$$\frac{2}{\eta_f} = \lambda_*.$$

Finally,

$$A_*^2 = \frac{3\lambda_*^2}{\Gamma_*}$$

by Proposition 5.6. Hence

$$B_*^2 = \lambda_*^2 A_*^2 \|v_*\|_\diamond^2 = \frac{3\lambda_*^4 \|v_*\|_\diamond^2}{\Gamma_*}.$$

□

Remark 6.2. The factorization

$$\mathcal{E}(\mu) = \sqrt{\mu} \beta(\mu)$$

is stronger than a formal leading-order expansion. It separates the universal square-root bifurcation scale from a regular geometric factor $\beta(\mu)$.

This form will allow the first-contact problem to be reduced to a regular implicit-function problem in the variable c^2 .

6.2 Monotonicity and uniqueness of the first contact

We next show that the exposure increases strictly along the newborn period-two branch.

Proposition 6.3 (Local monotonicity of the exposure). *There exists $\mu_1 > 0$ such that*

$$\frac{d}{d\mu} \mathcal{E}(\mu) > 0 \quad \text{for all } 0 < \mu < \mu_1. \tag{6.7}$$

Consequently, $\mathcal{E}$ is strictly increasing on $(0, \mu_1)$.

Proof. By Lemma 6.1,

$$\mathcal{E}(\mu) = \sqrt{\mu} \beta(\mu),$$

where

$$\beta(0) = B_* > 0.$$

For $\mu > 0$,

$$\mathcal{E}'(\mu) = \frac{\beta(\mu)}{2\sqrt{\mu}} + \sqrt{\mu} \beta'(\mu). \tag{6.8}$$

Since β is continuous and $\beta(0) = B_* > 0$, there exists $\mu_1 > 0$ such that

$$\beta(\mu) \geq \frac{B_*}{2}$$

for $0 \leq \mu < \mu_1$. Moreover, β' is bounded on this interval. Therefore the first term in (6.8), which is of order $\mu^{-1/2}$, dominates the second term for sufficiently small $\mu > 0$.

Hence

$$\mathcal{E}'(\mu) > 0$$

after reducing μ_1 if necessary. □

We may therefore define the first-contact parameter unambiguously.

Definition 6.4 (First switching-contact parameter). For sufficiently small $c > 0$, let $\mu_{\text{sc}}(c)$ denote the unique positive solution of

$$\mathcal{E}(\mu_{\text{sc}}(c)) = c. \quad (6.9)$$

The corresponding step size is

$$\eta_{\text{sc}}(c) := \eta_f + \mu_{\text{sc}}(c). \quad (6.10)$$

6.3 The quadratic separation law

We now state the central result of the paper.

Theorem 6.5 (Flip-to-switching separation law). *Assume (H1)–(H5). There exists $c_0 > 0$ such that for every*

$$0 < c < c_0$$

the first-contact parameter $\eta_{\text{sc}}(c)$ is well defined and unique.

As $c \rightarrow 0^+$,

$$\boxed{\eta_{\text{sc}}(c) - \eta_f = K_{\diamond} c^2 + O(c^4)} \quad (6.11)$$

where

$$\boxed{K_{\diamond} = \frac{1}{\lambda_*^2 A_*^2 \|v_*\|_{\diamond}^2}} \quad (6.12)$$

or, equivalently,

$$\boxed{K_{\diamond} = \frac{\ell_*}{\lambda_*^3 \|v_*\|_{\diamond}^2}} \quad (6.13)$$

and

$$\boxed{K_{\diamond} = \frac{\Gamma_*}{3\lambda_*^4 \|v_*\|_{\diamond}^2}}. \quad (6.14)$$

In particular,

$$\lim_{c \rightarrow 0^+} \frac{\eta_{\text{sc}}(c) - \eta_f}{c^2} = K_{\diamond} > 0. \quad (6.15)$$

Proof. By Proposition 6.3, the exposure $E(\mu)$ is strictly increasing for all sufficiently small $\mu > 0$. Since

$$E(\mu) \rightarrow 0 \quad \text{as} \quad \mu \rightarrow 0^+,$$

continuity and strict monotonicity imply that, for every sufficiently small $c > 0$, there exists a unique $\mu_{\text{sc}}(c) > 0$ satisfying

$$E(\mu_{\text{sc}}(c)) = c.$$

By Theorem 5.8,

$$E(\mu) = B_* \sqrt{\mu} + O(\mu^{3/2}) = \sqrt{\mu} (B_* + O(\mu)).$$

Therefore, at $\mu = \mu_{\text{sc}}(c)$,

$$c^2 = \mu_{\text{sc}}(c) (B_* + O(\mu_{\text{sc}}(c)))^2,$$

and hence

$$c^2 = B_*^2 \mu_{\text{sc}}(c) + O(\mu_{\text{sc}}(c)^2).$$

Since

$$\frac{E(\mu)}{\sqrt{\mu}} \rightarrow B_* > 0,$$

there exists $\mu_0 > 0$ such that

$$E(\mu) \geq \frac{B_*}{2} \sqrt{\mu}$$

for all $0 < \mu < \mu_0$. Evaluating this inequality at $\mu = \mu_{\text{sc}}(c)$ gives

$$\mu_{\text{sc}}(c) = O(c^2).$$

Substituting this estimate into the preceding relation yields

$$c^2 = B_*^2 \mu_{\text{sc}}(c) + O(c^4).$$

Consequently,

$$\mu_{\text{sc}}(c) = \frac{1}{B_*^2} c^2 + O(c^4).$$

Since

$$\eta_{\text{sc}}(c) = \eta_f + \mu_{\text{sc}}(c),$$

we obtain

$$\eta_{\text{sc}}(c) - \eta_f = K_\diamond c^2 + O(c^4),$$

where

$$K_\diamond = \frac{1}{B_*^2}.$$

Using

$$B_* = \lambda_* A_* \|v_*\|_\diamond$$

gives

$$K_\diamond = \frac{1}{\lambda_*^2 A_*^2 \|v_*\|_\diamond^2}.$$

Since

$$A_*^2 = \frac{\lambda_*}{\ell_*},$$

we further obtain

$$K_\diamond = \frac{\ell_*}{\lambda_*^3 \|v_*\|_\diamond^2}.$$

Finally,

$$\ell_* = \frac{\Gamma_*}{3\lambda_*},$$

and therefore

$$K_\diamond = \frac{\Gamma_*}{3\lambda_*^4 \|v_*\|_\diamond^2}.$$

By (H4), $\Gamma_* > 0$, so

$$K_\diamond > 0.$$

□

6.4 Explicit tensor form of the separation coefficient

The coefficient in Theorem 6.5 can be written entirely in terms of the local differential geometry of the potential.

Corollary 6.6 (Explicit separation coefficient). *Under the assumptions of Theorem 6.5,*

$$K_\diamond = \frac{3T_*[v_*, v_*, H_*^{-1}g_*] - Q_*[v_*, v_*, v_*, v_*]}{3\lambda_*^4 \|v_*\|_\diamond^2}, \quad (6.16)$$

where

$$g_* = T_*[v_*, v_*, \cdot]^\sharp.$$

Equivalently, in an orthonormal Hessian eigenbasis $\{v_i\}_{i=1}^d$,

$$K_\diamond = \frac{1}{3\lambda_*^4 \|v_*\|_\diamond^2} \left[3 \sum_{i=1}^d \frac{T_*[v_*, v_*, v_i]^2}{\lambda_i} - Q_*[v_*, v_*, v_*, v_*] \right]. \quad (6.17)$$

Proof. Substitute the representations (2.22) and (2.23) into (6.14). □

Remark 6.7 (Geometric interpretation). The coefficient $K_\diamond$ combines three distinct pieces of local geometry.

First,

$$\lambda_*$$

determines the linear flip threshold and the leading displacement of the gradient in the critical direction.

Second,

$$\Gamma_*$$

determines the nonlinear saturation of the unstable critical mode and hence the amplitude of the emerging period-two orbit.

Third,

$$\|v_*\|_\diamond$$

measures the critical eigendirection in the norm used by the clipping rule.

Thus the switching-contact scale is determined jointly by the smooth bifurcation geometry and the geometry of the clipping norm.

6.5 Euclidean gradient clipping

For standard gradient clipping,

$$\|\cdot\|_\diamond = \|\cdot\|_2.$$

Since

$$\|v_*\|_2 = 1,$$

the separation law simplifies substantially.

Corollary 6.8 (Euclidean clipping). *For the standard Euclidean clipping rule,*

$$\boxed{\eta_{\text{sc}}(c) - \frac{2}{\lambda_*} = K_2 c^2 + O(c^4),} \quad (6.18)$$

where

$$\boxed{K_2 = \frac{\ell_*}{\lambda_*^3} = \frac{\Gamma_*}{3\lambda_*^4}.} \quad (6.19)$$

Equivalently,

$$\boxed{K_2 = \frac{1}{3\lambda_*^4} \left[3 \sum_{i=1}^d \frac{T_*[v_*, v_*, v_i]^2}{\lambda_i} - Q_*[v_*, v_*, v_*, v_*] \right].} \quad (6.20)$$

6.6 The same coefficient controls two phenomena

The separation coefficient admits a direct interpretation in terms of the smooth flip coefficient.

Corollary 6.9 (Flip-switching coefficient relation). *The first flip coefficient ℓ_* and the leading switching-separation coefficient satisfy*

$$\boxed{K_\diamond = \frac{\ell_*}{\lambda_*^3 \|v_*\|_\diamond^2}.} \quad (6.21)$$

Therefore, within the present gradient structure, the same nonlinear coefficient that determines the birth and stability of the smooth period-two branch also determines its leading-order distance from the clipping boundary in parameter space.

Remark 6.10 (Interpretation for Euclidean clipping). For Euclidean clipping,

$$K_2 = \frac{\ell_*}{\lambda_*^3}.$$

Hence a larger positive supercritical flip coefficient produces a smaller initial oscillation amplitude,

$$A_*^2 = \frac{\lambda_*}{\ell_*},$$

and therefore delays the first clipping contact to a larger value of $\eta - \eta_f$.

Conversely, a smaller positive ℓ_* produces a larger small-amplitude cycle for the same μ and causes clipping to be encountered sooner.

6.7 Dynamical regimes around the first contact

The first-contact parameter separates two distinct local regimes.

Proposition 6.11 (Pre-contact and contact regimes). *There exist $c_0 > 0$ and $\mu_0 > 0$ such that for every $0 < c < c_0$ the following statements hold.*

(i) *If*

$$0 < \mu < \mu_{\text{sc}}(c),$$

then

$$x_{\pm}(\mu) \in \Omega_c^-. \quad (6.22)$$

Consequently, $\mathcal{O}_2(\mu)$ is simultaneously a stable period-two cycle of the smooth map $G_{\eta_f+\mu}$ and of the clipped map $F_{\eta_f+\mu,c}$.

(ii) *At*

$$\mu = \mu_{\text{sc}}(c),$$

both points satisfy

$$x_+(\mu_{\text{sc}}) \in \Sigma_c, \quad x_-(\mu_{\text{sc}}) \in \Sigma_c. \quad (6.23)$$

The same two points still satisfy the period-two equations for $F_{\eta_{\text{sc}},c}$.

(iii) *If*

$$\mu_{\text{sc}}(c) < \mu < \mu_0,$$

then the continuation of the smooth reference branch satisfies

$$x_{\pm}(\mu) \in \Omega_c^+. \quad (6.24)$$

In general, however, this smooth reference cycle is no longer a period-two cycle of the clipped map.

Proof. By Proposition 6.3, the exposure $E(\mu)$ is strictly increasing for all sufficiently small $\mu > 0$. We choose $\mu_0 > 0$ small enough that this monotonicity holds throughout $(0, \mu_0)$.

If

$$0 < \mu < \mu_{\text{sc}}(c),$$

then

$$E(\mu) < E(\mu_{\text{sc}}(c)) = c.$$

By Theorem 5.10, both points therefore lie in the smooth region:

$$x_{\pm}(\mu) \in \Omega_c^-.$$

Since $F_{\eta,c}$ and G_{η} coincide at both cycle points, the smooth period-two cycle is also a period-two cycle of the clipped map.

At

$$\mu = \mu_{\text{sc}}(c),$$

we have

$$E(\mu_{\text{sc}}(c)) = c.$$

Hence

$$x_+(\mu_{\text{sc}}) \in \Sigma_c, \quad x_-(\mu_{\text{sc}}) \in \Sigma_c.$$

Moreover, the clipping operator satisfies

$$C_c^{\diamond}(g) = g \quad \text{whenever } \|g\|_{\diamond} = c.$$

Thus the clipped and smooth maps still agree at both cycle points, and the period-two relations remain valid at the contact parameter.

Finally, if

$$\mu_{\text{sc}}(c) < \mu < \mu_0,$$

strict monotonicity gives

$$E(\mu) > c.$$

Therefore

$$x_{\pm}(\mu) \in \Omega_c^+.$$

These points still form the continuation of the smooth reference cycle, but in general they no longer form a period-two cycle of the clipped map, because in Ω_c^+ the gradient update is rescaled by the factor

$$\frac{c}{\|\nabla V(x)\|_{\diamond}} < 1.$$

□

Remark 6.12 (Stability at the contact point). At the exact switching-contact parameter, the two-cycle continues to exist as an orbit of the clipped map because the map itself is continuous across Σ_c .

We do not, however, infer its stability directly from the smooth multiplier formula. The derivative of the clipped map has different one-sided limits at the switching surface, so stability at and beyond contact is a genuinely piecewise-smooth problem.

This issue is deferred to the post-contact analysis.

6.8 Origin of the improved remainder

The $O(c^4)$ remainder in Theorem 6.5 is a structural consequence of the exact gradient-balance identity.

Indeed, Theorem 5.8 gives

$$E(\mu) = B_*\sqrt{\mu} + O(\mu^{3/2}),$$

rather than the more general expansion

$$E(\mu) = B_1\sqrt{\mu} + B_2\mu + O(\mu^{3/2}).$$

The absence of the $O(\mu)$ term is a consequence of the oddness of the half-difference in the signed square-root parameter and of the exact identity

$$E(\mu) = \frac{2}{\eta_f + \mu} \|d(\sqrt{\mu})\|_{\diamond}.$$

Squaring the actual exposure expansion gives

$$E(\mu)^2 = B_*^2\mu + O(\mu^2).$$

At first contact,

$$E(\mu_{\text{sc}}(c)) = c,$$

and therefore

$$c^2 = B_*^2\mu_{\text{sc}}(c) + O(\mu_{\text{sc}}(c)^2).$$

Since

$$\mu_{\text{sc}}(c) = O(c^2),$$

it follows that

$$\mu_{\text{sc}}(c) = \frac{1}{B_*^2}c^2 + O(c^4).$$

Thus the improvement from the naively expected $O(c^3)$ remainder to $O(c^4)$ does not require a higher-order normal-form coefficient. It follows directly from the exact two-cycle gradient balance and the resulting parity structure of the exposure.

A further expansion of the form

$$\eta_{\text{sc}}(c) = \eta_f + K_{\diamond}c^2 + K_{\diamond,4}c^4 + o(c^4)$$

would require additional higher-order information on the period-two branch. Such coefficients are not needed for the leading separation law considered here.

7 Post-contact degeneracy and fully clipped period-two dynamics

Sections 5 and 6 show that the smooth period-two branch created by the supercritical flip reaches the switching hypersurface simultaneously at both of its points.

The radial clipping structure imposes a further restriction: a genuine period-two orbit cannot have one point in the smooth region and the other in the clipped region. Hence any period-two orbit that persists beyond the first switching contact must enter the clipped regime at both points.

The post-contact results obtained below are not consequences of the generic border-collision theory cited in Section 1. They rely specifically on the radial structure of the clipping operator.

The purpose of this section is to characterize the universal structure of such fully clipped period-two orbits. The main result is that every fully clipped period-two orbit is necessarily nonhyperbolic: its two-step monodromy matrix has an exact multiplier equal to one.

This neutral multiplier is not a consequence of the local flip normal form. It arises directly from the radial normalization inherent in gradient clipping.

7.1 Fully clipped period-two orbits

We first record the exact algebraic structure of a fully clipped period-two orbit.

Proposition 7.1 (Exact geometry of a fully clipped two-cycle). *Let*

$$\mathcal{Y}_2 = \{y_-, y_+\}$$

be a genuine period-two orbit of

$$F_{\eta,c}(x) = x - \eta \mathcal{C}_c^\diamond(\nabla V(x)),$$

and suppose

$$y_-, y_+ \in \Omega_c^+.$$

Then

$$\boxed{\|y_+ - y_-\|_\diamond = \eta c.} \tag{7.1}$$

Moreover, there exist

$$u \in \mathbb{R}^d, \quad \|u\|_\diamond = 1,$$

and numbers

$$r_+, r_- > c$$

such that

$$y_+ - y_- = \eta c u, \tag{7.2}$$

$$\nabla V(y_+) = r_+ u, \tag{7.3}$$

$$\nabla V(y_-) = -r_- u. \tag{7.4}$$

Proof. The period-two relations are

$$y_- = y_+ - \eta \mathcal{C}_c^\diamond(\nabla V(y_+)), \tag{7.5}$$

$$y_+ = y_- - \eta \mathcal{C}_c^\diamond(\nabla V(y_-)). \tag{7.6}$$

Therefore

$$\mathcal{C}_c^\diamond(\nabla V(y_+)) = \frac{y_+ - y_-}{\eta}, \tag{7.7}$$

$$\mathcal{C}_c^\diamond(\nabla V(y_-)) = -\frac{y_+ - y_-}{\eta}. \tag{7.8}$$

Since both points lie in the strictly clipped region,

$$\|\mathcal{C}_c^\diamond(\nabla V(y_\pm))\|_\diamond = c.$$

Taking norms in (7.7) gives

$$\|y_+ - y_-\|_\diamond = \eta c.$$

Define

$$u = \frac{y_+ - y_-}{\eta c}.$$

Then

$$\|u\|_\diamond = 1.$$

Because radial clipping preserves direction,

$$\frac{\nabla V(y_+)}{\|\nabla V(y_+)\|_\diamond} = u$$

and

$$\frac{\nabla V(y_-)}{\|\nabla V(y_-)\|_\diamond} = -u.$$

Setting

$$r_\pm = \|\nabla V(y_\pm)\|_\diamond$$

gives the result. $\square$

Remark 7.2 (Amplitude saturation). Before contact, the period-two amplitude is generated by the smooth flip mechanism and satisfies

$$\|x_+(\mu) - x_-(\mu)\| = O(\sqrt{\mu}).$$

After both points enter the clipped regime, the step length is no longer determined by the smooth normal form. Instead, radial clipping imposes the exact constraint

$$\|y_+ - y_-\|_\diamond = \eta c.$$

Thus clipping replaces the smooth square-root amplitude law by an exact saturated step-length law.

7.2 Exclusion of mixed period-two itineraries

We now state explicitly the structural restriction established in Section 5.

Proposition 7.3 (No mixed smooth-clipped period-two orbit). *A radial clipped gradient map cannot possess a genuine period-two orbit*

$$\{y_-, y_+\}$$

such that exactly one point lies in Ω_c^- and the other lies in Ω_c^+ .

Consequently, every genuine period-two orbit is locally of one of the following types:

- (i) *both points are smooth;*
- (ii) *both points lie on the switching boundary;*
- (iii) *both points are strictly clipped.*

Proof. Suppose, for contradiction, that

$$y_- \in \Omega_c^-, \quad y_+ \in \Omega_c^+.$$

Then the period-two equations imply

$$\nabla V(y_-) + \mathcal{C}_c^\diamond(\nabla V(y_+)) = 0.$$

Taking the clipping norm gives

$$\|\nabla V(y_-)\|_\diamond = \|\mathcal{C}_c^\diamond(\nabla V(y_+))\|_\diamond = c,$$

which contradicts

$$\|\nabla V(y_-)\|_\diamond < c.$$

The opposite mixed itinerary is ruled out identically. $\square$

Remark 7.4. Proposition 7.3 is not a local consequence of the flip bifurcation. It follows from the algebraic structure of radial clipping and therefore applies to arbitrary period-two orbits, independently of their amplitude or origin.

7.3 Differential structure in the clipped region

We now examine the differential structure of the radial clipping operator in the strictly clipped region. This structure will be the key ingredient in the spectral analysis of fully clipped period-two orbits.

Let

$$\rho(g) := \|g\|_{\diamond}.$$

For any nonzero vector $g \in \mathbb{R}^d$, write

$$u := \frac{g}{\rho(g)}, \quad \rho(u) = 1,$$

and define the covector

$$\phi_u := D\rho(u).$$

Since ρ is positively homogeneous of degree one, Euler's identity gives

$$\phi_u[u] = D\rho(u)[u] = \rho(u) = 1. \quad (7.9)$$

Moreover, the derivative of a positively homogeneous function of degree one is homogeneous of degree zero. Hence, since $g = \rho(g)u$ with $\rho(g) > 0$,

$$D\rho(g) = D\rho(u) = \phi_u. \quad (7.10)$$

In the strictly clipped region

$$\rho(g) > c,$$

the clipping operator takes the form

$$C_c^\diamond(g) = c \frac{g}{\rho(g)}.$$

For any $h \in \mathbb{R}^d$, differentiation yields

$$DC_c^\diamond(g)[h] = \frac{c}{\rho(g)} h - \frac{c}{\rho(g)^2} g D\rho(g)[h].$$

Using

$$g = \rho(g)u$$

and (7.10), we obtain

$$DC_c^\diamond(g)[h] = \frac{c}{\rho(g)} (h - u \phi_u[h]).$$

Equivalently,

$$DC_c^\diamond(g) = \frac{c}{\rho(g)} (I - u \otimes \phi_u). \quad (7.11)$$

The operator

$$P_u := I - u \otimes \phi_u$$

has a simple geometric interpretation. By (7.9),

$$P_u u = u - u \phi_u[u] = 0,$$

and

$$\phi_u P_u = \phi_u - \phi_u[u] \phi_u = 0.$$

Thus P_u projects onto

$$T_u\{\rho = 1\} = \ker \phi_u,$$

the tangent space of the unit ρ -sphere at u , along the radial direction spanned by u .

In particular, multiplying (7.11) from the left by ϕ_u gives

$$\phi_u DC_c^\diamond(g) = \frac{c}{\rho(g)} (\phi_u - \phi_u[u] \phi_u) = 0. \quad (7.12)$$

Equation (7.12) expresses the differential degeneracy created by radial clipping: infinitesimal variations in the gradient-magnitude direction are annihilated, whereas only variations tangent to the clipping sphere contribute to the differential of the clipped update.

For every $x \in \Omega_c^+$, the chain rule therefore gives

$$DF_{\eta,c}(x) = I - \eta DC_c^\diamond(\nabla V(x)) D^2 V(x). \quad (7.13)$$

Proposition 7.5 (Pointwise neutral covector). *Let $x \in \Omega_c^+$ and set*

$$u = \frac{\nabla V(x)}{\|\nabla V(x)\|_\diamond}.$$

Then

$$\phi_u DF_{\eta,c}(x) = \phi_u. \quad (7.14)$$

In particular,

$$1 \in \sigma(DF_{\eta,c}(x)).$$

Proof. Using (7.13),

$$\phi_u DF_{\eta,c}(x) = \phi_u - \eta \phi_u DC_c^\diamond(\nabla V(x)) D^2 V(x).$$

By (7.12),

$$\phi_u DC_c^\diamond(\nabla V(x)) = 0.$$

Hence the second term vanishes and

$$\phi_u DF_{\eta,c}(x) = \phi_u.$$

Thus ϕ_u is a left eigenvector of $DF_{\eta,c}(x)$ associated with the eigenvalue 1. Therefore

$$1 \in \sigma(DF_{\eta,c}(x)).$$

□

Remark 7.6 (Pointwise neutrality versus orbit neutrality). Although every one-step Jacobian in the strictly clipped region has a unit eigenvalue, the corresponding left eigenvector depends on the local gradient direction. Hence, along a general clipped orbit, unit eigenvalues of the individual Jacobians do not imply that their product possesses a unit Floquet multiplier.

Period-two orbits are exceptional. As shown in Proposition 7.1, the clipped period-two equations force the two gradient directions to be exactly opposite. Moreover, because ρ is even,

$$\rho(-u) = \rho(u),$$

and therefore

$$D\rho(-u) = -D\rho(u).$$

Consequently,

$$(-u) \otimes D\rho(-u) = u \otimes D\rho(u),$$

so the tangent projection

$$I - u \otimes \phi_u$$

is unchanged under $u \mapsto -u$.

Thus the same neutral covector structure is preserved at both points of a fully clipped period-two orbit. This observation is the mechanism behind the universal unit-multiplier theorem proved in the next subsection.

7.4 Universal unit multiplier for fully clipped two-cycles

We now obtain the principal post-contact result.

Theorem 7.7 (Universal unit multiplier). *Let*

$$\mathcal{Y}_2 = \{y_-, y_+\}$$

be any fully clipped period-two orbit of $F_{\eta,c}$.

Let

$$M_c := DF_{\eta,c}(y_-) DF_{\eta,c}(y_+)$$

be the corresponding two-step monodromy matrix.

Then

$$\boxed{1 \in \sigma(M_c)}. \quad (7.15)$$

More precisely, if u is the common clipping direction defined by Proposition 7.1, then

$$\varphi_u DF_{\eta,c}(y_+) = \varphi_u \quad (7.16)$$

and

$$\varphi_u DF_{\eta,c}(y_-) = \varphi_u. \quad (7.17)$$

Hence

$$\varphi_u M_c = \varphi_u. \quad (7.18)$$

Proof. By Proposition 7.1, there exist $r_+, r_- > c$ and a vector u with $\|u\|_\diamond = 1$ such that

$$\nabla V(y_+) = r_+ u, \quad \nabla V(y_-) = -r_- u.$$

Since every norm is even,

$$\rho(-z) = \rho(z),$$

and therefore

$$D\rho(-u) = -D\rho(u).$$

Thus

$$\phi_{-u} = -\phi_u.$$

At y_+ , the normalized gradient direction is u . Hence Proposition 7.5 gives

$$\phi_u DF_{\eta,c}(y_+) = \phi_u.$$

At y_- , the normalized gradient direction is $-u$. Applying Proposition 7.5 with $-u$ gives

$$\phi_{-u} DF_{\eta,c}(y_-) = \phi_{-u}.$$

Since $\phi_{-u} = -\phi_u$, this is equivalent to

$$\phi_u DF_{\eta,c}(y_-) = \phi_u.$$

Consequently,

$$\phi_u M_c = \phi_u DF_{\eta,c}(y_-) DF_{\eta,c}(y_+) = \phi_u.$$

Thus ϕ_u is a nonzero left eigenvector of M_c associated with the eigenvalue 1. Therefore

$$1 \in \sigma(M_c).$$

□

Corollary 7.8 (Fully clipped period-two cycles are nonhyperbolic). *Every fully clipped period-two orbit of a radial clipped gradient map is nonhyperbolic.*

Proof. Theorem 7.7 gives a multiplier equal to 1, which lies on the unit circle. □

Corollary 7.9 (No linear exponential attraction). *A fully clipped period-two orbit cannot be linearly exponentially asymptotically stable.*

Proof. Linear exponential asymptotic stability requires every Floquet multiplier of the period-two orbit to lie strictly inside the unit disk. Theorem 7.7 provides an exact multiplier equal to 1. □

Remark 7.10 (Nonhyperbolic does not mean unstable). The presence of the unit multiplier does not by itself imply nonlinear instability.

A fully clipped period-two orbit may be nonlinearly attracting, repelling, neutral, or embedded in a family of periodic orbits. These alternatives depend on higher-order dynamics in the neutral direction and cannot be distinguished by the linearization alone.

7.5 The spectral transition at first contact

Before contact, the smooth period-two orbit is hyperbolically attracting along its critical direction.

By Theorem 4.5,

$$\rho_{\text{sm}}^c(\mu) = 1 - 4\lambda_*\mu + O(\mu^{3/2}). \quad (7.19)$$

At the first switching contact,

$$\mu_{\text{sc}}(c) = K_{\diamond}c^2 + O(c^4). \quad (7.20)$$

Hence

$$\rho_{\text{sm}}^c(\mu_{\text{sc}}(c)) = 1 - 4\lambda_*K_{\diamond}c^2 + O(c^3). \quad (7.21)$$

Since

$$K_{\diamond} = \frac{\Gamma_*}{3\lambda_*^4\|v_*\|_{\diamond}^2},$$

this becomes

$$\rho_{\text{sm}}^c(\mu_{\text{sc}}(c)) = 1 - \frac{4\Gamma_*}{3\lambda_*^3\|v_*\|_{\diamond}^2}c^2 + O(c^3). \quad (7.22)$$

Thus the smooth cycle reaches the switching boundary while still weakly attracting:

$$\rho_{\text{sm}}^c < 1.$$

Any fully clipped period-two orbit beyond contact, however, satisfies

$$\rho_{\text{clip}}^c = 1$$

in at least one direction.

This yields the following conclusion.

Corollary 7.11 (Loss of hyperbolicity under fully clipped continuation). *Suppose that a period-two orbit admits a continuation from the first switching contact into the fully clipped regime. Then every such fully clipped period-two continuation is necessarily nonhyperbolic.*

Immediately before contact, the smooth critical multiplier satisfies

$$\rho_{\text{sm}}^c = 1 - 4\lambda_*K_{\diamond}c^2 + O(c^3),$$

and therefore

$$1 - \rho_{\text{sm}}^c = 4\lambda_*K_{\diamond}c^2 + O(c^3).$$

Thus the smooth period-two branch reaches the switching boundary while still weakly attracting, whereas, by Theorem 7.7, every fully clipped period-two orbit carries an exact unit multiplier.

Remark 7.12. The transition should not be interpreted as a regular perturbation of the smooth two-cycle multiplier.

The derivative of radial clipping changes discontinuously across the switching boundary. On the clipped side, radial variations of the gradient are annihilated, and this rank-deficient structure produces an exact neutral direction.

Thus the post-contact spectrum is constrained by the geometry of the clipping operator rather than obtained by smoothly perturbing the pre-contact smooth monodromy.

7.6 Degeneracy of the period-two continuation equation

A period-two point of the clipped map is a solution of

$$\mathcal{P}(y, \eta, c) = F_{\eta, c}^2(y) - y = 0. \quad (7.23)$$

If the two-cycle were hyperbolic, one would normally attempt to continue it by applying the implicit function theorem to

$$\mathcal{P} = 0.$$

The universal unit multiplier prevents this.

Proposition 7.13 (Singularity of the post-contact continuation problem). *Let y belong to a fully clipped period-two orbit. Then*

$$D_y \mathcal{P}(y, \eta, c) = DF_{\eta, c}^2(y) - I \quad (7.24)$$

is singular.

Consequently, assumptions (H1)–(H5) do not yield a standard implicit-function continuation of the period-two branch through the fully clipped regime.

Proof. By Theorem 7.7,

$$1 \in \sigma \left(DF_{\eta, c}^2(y) \right).$$

Hence

$$0 \in \sigma \left(DF_{\eta, c}^2(y) - I \right),$$

so the derivative of the period-two fixed-point equation is singular. $\square$

Remark 7.14 (Why a universal post-contact branch theorem is absent). The smooth period-two branch before contact is governed by a nondegenerate flip bifurcation and therefore admits a standard local branch description.

After clipping becomes active, the relevant periodic-orbit equation is intrinsically degenerate. Additional nonlinear information is required to determine whether the fully clipped periodic orbit:

- persists as an isolated nonhyperbolic two-cycle;
- belongs to a family of two-cycles;
- is nonlinearly attracting or repelling;
- or terminates at the switching contact.

The solvable planar model of Section 8 realizes the second possibility: the post-contact dynamics contain a one-parameter family of fully clipped period-two cycles.

7.7 Euclidean clipping

For the Euclidean norm,

$$\| \cdot \|_{\diamond} = \| \cdot \|_2,$$

the geometry becomes especially transparent.

If

$$u = \frac{\nabla V(x)}{\|\nabla V(x)\|_2},$$

then in the clipped region

$$D\mathcal{C}_c(\nabla V(x)) = \frac{c}{\|\nabla V(x)\|_2} (I - uu^{\top}). \quad (7.25)$$

Hence

$$DF_{\eta, c}(x) = I - \frac{\eta c}{\|\nabla V(x)\|_2} (I - uu^{\top}) D^2 V(x). \quad (7.26)$$

The radial covector satisfies

$$u^{\top} DF_{\eta, c}(x) = u^{\top}. \quad (7.27)$$

For a fully clipped period-two orbit with gradient directions $\pm u$,

$$u^{\top} DF_{\eta, c}(y_-) DF_{\eta, c}(y_+) = u^{\top}. \quad (7.28)$$

Thus the unit multiplier has a direct geometric interpretation: Euclidean clipping removes the derivative of the update in the radial gradient-magnitude direction.

7.8 Summary of the post-contact mechanism

The local transition established in this paper can therefore be summarized as follows.

For

$$0 < \eta - \eta_f < \eta_{\text{sc}}(c) - \eta_f,$$

the period-two orbit lies entirely in the smooth region and is governed by the supercritical smooth flip.

At

$$\eta = \eta_{\text{sc}}(c),$$

both points of the orbit reach the switching boundary simultaneously.

A mixed smooth–clipped two-cycle cannot occur.

Any genuine period-two continuation into the clipped region must satisfy

$$\|y_+ - y_-\|_\diamond = \eta c$$

and necessarily possesses an exact unit multiplier:

$$1 \in \sigma(M_c).$$

Hence the first clipping contact marks a transition

$$\begin{array}{ccc} \text{smooth attracting two-cycle} & & \\ \Downarrow & & \\ \text{simultaneous clipping contact} & & (7.29) \\ \Downarrow & & \\ \text{nonhyperbolic clipped regime.} & & \end{array}$$

This degeneracy is a structural consequence of radial clipping and does not depend on the specific solvable model considered later.

8 A completely solvable planar gradient model

We now present a two-dimensional model for which the smooth flip bifurcation, the first clipping contact, the separation coefficient, and the post-contact dynamics can all be computed explicitly.

Consider the potential

$$V(x, y) = \frac{\lambda}{2}x^2 + \frac{\nu}{2}y^2 - \frac{b}{4}x^4 + \frac{d}{6}x^6, \quad (8.1)$$

where

$$\lambda > \nu > 0, \quad b > 0, \quad d > 0, \quad b^2 < 4d\lambda. \quad (8.2)$$

We use the standard Euclidean clipping norm throughout this section.

The gradient is

$$\nabla V(x, y) = \begin{pmatrix} \lambda x - bx^3 + dx^5 \\ \nu y \end{pmatrix}. \quad (8.3)$$

For convenience, define

$$g(x) := \lambda x - bx^3 + dx^5 = x(\lambda - bx^2 + dx^4). \quad (8.4)$$

The condition $b^2 < 4d\lambda$ implies

$$\lambda - bs + ds^2 > 0 \quad \text{for every } s \geq 0, \quad (8.5)$$

and hence

$$xg(x) > 0 \quad \text{for every } x \neq 0. \quad (8.6)$$

Thus $(0, 0)$ is the unique critical point of V and is a strict global minimum.

8.1 Verification of the standing hypotheses

At the origin,

$$H_* = D^2V(0, 0) = \begin{pmatrix} \lambda & 0 \\ 0 & \nu \end{pmatrix}. \quad (8.7)$$

Since

$$\lambda > \nu > 0,$$

the maximal Hessian eigenvalue is

$$\lambda_* = \lambda \quad (8.8)$$

and its normalized eigenvector is

$$v_* = e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \quad (8.9)$$

The flip threshold is therefore

$$\boxed{\eta_f = \frac{2}{\lambda}}. \quad (8.10)$$

Since the potential is even in x ,

$$D^3V(0, 0) = 0. \quad (8.11)$$

Moreover,

$$D^4V(0, 0)[e_1, e_1, e_1, e_1] = -6b. \quad (8.12)$$

Hence

$$\Gamma_* = -D^4V(0, 0)[e_1^4] = 6b > 0. \quad (8.13)$$

Thus the flip is supercritical.

The first flip coefficient is

$$\boxed{\ell_* = \frac{\Gamma_*}{3\lambda} = \frac{2b}{\lambda}}. \quad (8.14)$$

The general amplitude formula gives

$$\boxed{A_*^2 = \frac{3\lambda^2}{\Gamma_*} = \frac{\lambda^2}{2b}}, \quad (8.15)$$

or

$$A_* = \frac{\lambda}{\sqrt{2b}}. \quad (8.16)$$

Finally, the general Euclidean separation coefficient becomes

$$\boxed{K_2 = \frac{\Gamma_*}{3\lambda^4} = \frac{2b}{\lambda^4}}. \quad (8.17)$$

We shall recover all of these quantities below from the exact dynamics.

8.2 Exact smooth gradient dynamics

The ordinary gradient-descent map is

$$G_\eta(x, y) = \begin{pmatrix} (1 - \eta\lambda)x + \eta bx^3 - \eta dx^5 \\ (1 - \eta\nu)y \end{pmatrix}. \quad (8.18)$$

The line

$$\mathcal{L} = \{(x, 0) : x \in \mathbb{R}\} \quad (8.19)$$

is invariant.

On $\mathcal{L}$, the dynamics reduce to the odd scalar map

$$f_\eta(x) = (1 - \eta\lambda)x + \eta bx^3 - \eta dx^5. \quad (8.20)$$

Because f_η is odd, a symmetric period-two orbit has the form

$$\{-a, a\}, \quad a > 0.$$

The condition

$$f_\eta(a) = -a$$

is equivalent to

$$\lambda - \frac{2}{\eta} = ba^2 - da^4. \quad (8.21)$$

Set

$$q := a^2. \quad (8.22)$$

Then

$$dq^2 - bq + \lambda - \frac{2}{\eta} = 0. \quad (8.23)$$

The small-amplitude root is therefore

$$q_{\text{sm}}(\eta) = \frac{b - \sqrt{b^2 - 4d\left(\lambda - \frac{2}{\eta}\right)}}{2d}. \quad (8.24)$$

Thus the smooth period-two branch is given explicitly by

$$\boxed{\mathcal{O}_{2,\text{sm}}(\eta) = \{(-a_{\text{sm}}(\eta), 0), (a_{\text{sm}}(\eta), 0)\}}, \quad (8.25)$$

where

$$a_{\text{sm}}(\eta) = \sqrt{q_{\text{sm}}(\eta)}. \quad (8.26)$$

8.3 Exact verification of the square-root law

Let

$$\mu = \eta - \frac{2}{\lambda}. \quad (8.27)$$

Then

$$\lambda - \frac{2}{\eta} = \frac{\lambda^2}{2}\mu - \frac{\lambda^3}{4}\mu^2 + O(\mu^3). \quad (8.28)$$

Solving (8.21) gives

$$a_{\text{sm}}(\eta)^2 = \frac{\lambda^2}{2b}\mu + \frac{\lambda^3(d\lambda - b^2)}{4b^3}\mu^2 + O(\mu^3). \quad (8.29)$$

Taking the square root yields

$$a_{\text{sm}}(\eta) = \frac{\lambda}{\sqrt{2b}}\mu^{1/2} \left[1 + \frac{\lambda(d\lambda - b^2)}{4b^2}\mu + O(\mu^2) \right]. \quad (8.30)$$

Hence

$$a_{\text{sm}}(\eta) = A_*\sqrt{\mu} + O(\mu^{3/2}), \quad (8.31)$$

with

$$A_* = \frac{\lambda}{\sqrt{2b}},$$

exactly as predicted by (8.16).

Because the model is reflection-symmetric, the midpoint of the smooth two-cycle remains exactly at the critical point:

$$\frac{(a_{\text{sm}}, 0) + (-a_{\text{sm}}, 0)}{2} = (0, 0). \quad (8.32)$$

8.4 Stability of the smooth period-two branch

The derivative of the scalar map is

$$f'_\eta(x) = 1 - \eta\lambda + 3\eta bx^2 - 5\eta dx^4. \quad (8.33)$$

Using the cycle equation (8.21), one obtains

$$f'_\eta(a) = -1 + 2\eta a^2 (b - 2da^2). \quad (8.34)$$

Since f'_η is even, the critical multiplier of the two-step map is

$$\rho_x^{\text{sm}} = [-1 + 2\eta a^2 (b - 2da^2)]^2. \quad (8.35)$$

For $\mu \rightarrow 0^+$,

$$\rho_x^{\text{sm}} = 1 - 4\lambda\mu + O(\mu^2), \quad (8.36)$$

in agreement with Theorem 4.5.

The transverse multiplier is

$$\rho_y^{\text{sm}} = (1 - \eta\nu)^2. \quad (8.37)$$

Since $0 < \nu < \lambda$, at $\eta = \eta_f$ one has $|1 - 2\nu/\lambda| < 1$, so the transverse multiplier stays inside the unit disk. Consequently, the smooth period-two cycle is locally asymptotically stable for all sufficiently small $\eta - \eta_f > 0$.

8.5 Exact gradient exposure

At the positive point of the cycle,

$$\nabla V(a_{\text{sm}}, 0) = \begin{pmatrix} g(a_{\text{sm}}) \\ 0 \end{pmatrix}.$$

By the exact period-two relation,

$$g(a_{\text{sm}}) = \frac{2a_{\text{sm}}}{\eta}. \quad (8.38)$$

Hence the common Euclidean gradient exposure is

$$\boxed{\mathcal{E}(\eta) = \frac{2a_{\text{sm}}(\eta)}{\eta}}. \quad (8.39)$$

Using (8.30),

$$\mathcal{E}(\mu) = \frac{\lambda^2}{\sqrt{2b}} \mu^{1/2} \left[1 + \frac{\lambda(d\lambda - 3b^2)}{4b^2} \mu + O(\mu^2) \right]. \quad (8.40)$$

In particular,

$$B_* = \frac{\lambda^2}{\sqrt{2b}}, \quad (8.41)$$

and therefore

$$\frac{1}{B_*^2} = \frac{2b}{\lambda^4}, \quad (8.42)$$

again agreeing with the general separation coefficient (8.17).

8.6 Exact switching-contact equation

The first contact occurs when

$$\frac{2a_{\text{sm}}(\eta)}{\eta} = c. \quad (8.43)$$

Hence at contact

$$a_{\text{sm}} = \frac{\eta c}{2}. \quad (8.44)$$

Substituting this into (8.21) gives the exact scalar equation

$$\Psi(\eta, c) := \lambda - \frac{2}{\eta} - \frac{b}{4}\eta^2 c^2 + \frac{d}{16}\eta^4 c^4 = 0. \quad (8.45)$$

At

$$(\eta, c) = \left(\frac{2}{\lambda}, 0 \right),$$

we have

$$\Psi \left(\frac{2}{\lambda}, 0 \right) = 0$$

and

$$\partial_\eta \Psi \left(\frac{2}{\lambda}, 0 \right) = \frac{\lambda^2}{2} > 0. \quad (8.46)$$

Thus the implicit function theorem gives a unique smooth first-contact curve

$$\eta = \eta_{\text{sc}}(c)$$

for sufficiently small c .

8.7 Exact verification of the separation law

Write

$$\eta_{\text{sc}}(c) = \frac{2}{\lambda} + K_2 c^2 + K_4 c^4 + O(c^6). \quad (8.47)$$

Substituting this expansion into (8.45) and comparing powers of c gives

$$\boxed{K_2 = \frac{2b}{\lambda^4}}, \quad (8.48)$$

and

$$\boxed{K_4 = \frac{6b^2}{\lambda^7} - \frac{2d}{\lambda^6}}. \quad (8.49)$$

Therefore

$$\boxed{\eta_{\text{sc}}(c) = \frac{2}{\lambda} + \frac{2b}{\lambda^4} c^2 + \left(\frac{6b^2}{\lambda^7} - \frac{2d}{\lambda^6} \right) c^4 + O(c^6)}. \quad (8.50)$$

The leading coefficient

$$\frac{2b}{\lambda^4}$$

coincides exactly with

$$\frac{\Gamma_*}{3\lambda_*^4} = \frac{6b}{3\lambda^4},$$

as predicted by Theorem 6.5.

This model therefore verifies not only the quadratic exponent but also the absence of a cubic correction.

8.8 The fully clipped symmetric branch

We now turn to the post-contact regime. Define

$$a_c(\eta) := \frac{\eta c}{2}, \quad (8.51)$$

and consider the pair

$$Y_\pm(\eta, c) = (\pm a_c(\eta), 0). \quad (8.52)$$

Its gradient magnitude is

$$g(a_c) = a_c (\lambda - b a_c^2 + d a_c^4), \quad (8.53)$$

and a direct calculation gives

$$g(a_c) - c = \frac{\eta c}{2} \Psi(\eta, c), \quad (8.54)$$

with Ψ as in (8.45). Hence $g(a_c) = c$ at $\eta = \eta_{\text{sc}}(c)$, and since $\partial_\eta \Psi(\eta_{\text{sc}}(c), c) > 0$ for all sufficiently small c ,

$$g(a_c) > c \quad \text{for } \eta > \eta_{\text{sc}}(c) \quad (8.55)$$

sufficiently close to the contact value. Both points $Y_{\pm}$ therefore lie in the fully clipped region. On the invariant axis, radial clipping acts as

$$F_{\eta,c}(x, 0) = (x - \eta c, 0) \quad (g(x) > c), \quad F_{\eta,c}(x, 0) = (x + \eta c, 0) \quad (|g(x)| > c, x < 0),$$

and since $2a_c = \eta c$,

$$F_{\eta,c}(a_c, 0) = (-a_c, 0), \quad F_{\eta,c}(-a_c, 0) = (a_c, 0). \quad (8.56)$$

Thus:

Proposition 8.1 (Explicit fully clipped period-two branch). *For every sufficiently small $c > 0$ and every $\eta > \eta_{\text{sc}}(c)$ sufficiently close to $\eta_{\text{sc}}(c)$,*

$$\mathcal{O}_{2,\text{clip}}(\eta, c) = \left\{ \left(-\frac{\eta c}{2}, 0 \right), \left(\frac{\eta c}{2}, 0 \right) \right\} \quad (8.57)$$

is a fully clipped period-two orbit.

Its amplitude satisfies exactly

$$2a_c = \eta c. \quad (8.58)$$

8.9 A continuum of clipped period-two cycles

The symmetric clipped cycle is not isolated.

For

$$\eta > \eta_{\text{sc}}(c),$$

we have

$$g(a_c) > c.$$

By continuity, there exists $\delta > 0$ such that

$$g(a_c + s) > c, \quad g(a_c - s) > c \quad (8.59)$$

for every

$$|s| < \delta.$$

Define

$$Y_+(s) = (a_c + s, 0), \quad (8.60)$$

$$Y_-(s) = (-a_c + s, 0). \quad (8.61)$$

After reducing δ if necessary,

$$a_c + s > 0, \quad -a_c + s < 0.$$

Since

$$Y_+(s) - Y_-(s) = (\eta c, 0),$$

radial clipping gives

$$F_{\eta,c}(Y_+(s)) = Y_-(s), \quad (8.62)$$

$$F_{\eta,c}(Y_-(s)) = Y_+(s). \quad (8.63)$$

We therefore obtain the following stronger result.

Theorem 8.2 (A local family of clipped period-two cycles). *For every sufficiently small $c > 0$ and $\eta > \eta_{\text{sc}}(c)$ sufficiently close to contact, there exists $\delta > 0$ such that*

$$\mathcal{O}_2(s) = \{(-a_c + s, 0), (a_c + s, 0)\}, \quad |s| < \delta, \quad (8.64)$$

is a one-parameter family of fully clipped period-two cycles.

Equivalently,

$$F_{\eta,c}^2(x, 0) = (x, 0) \quad (8.65)$$

on a nontrivial interval of the positive clipped component and on its image under $F_{\eta,c}$.

Proof. For $|s| < \delta$, both points lie in the clipped regime by (8.59).

At $Y_+(s)$, the gradient points in the positive x direction, so the clipped step is exactly

$$-\eta c e_1.$$

Therefore

$$F_{\eta,c}(Y_+(s)) = (a_c + s - \eta c, 0) = (-a_c + s, 0) = Y_-(s).$$

At $Y_-(s)$, the gradient points in the negative x direction, so the clipped step is

$$+\eta c e_1.$$

Thus

$$F_{\eta,c}(Y_-(s)) = (-a_c + s + \eta c, 0) = (a_c + s, 0) = Y_+(s).$$

□

Remark 8.3 (Geometric meaning of the unit multiplier). The continuum in Theorem 8.2 gives a concrete nonlinear interpretation of the universal unit multiplier proved in Theorem 7.7.

The neutral multiplier is not merely an artifact of linearization: in this model the two-step map is exactly the identity along the one-dimensional clipped family.

8.10 Exact post-contact multipliers

Consider the period-two cycle

$$\mathcal{O}_2(s) = \{Y_-(s), Y_+(s)\}.$$

Set

$$r_+(s) = g(a_c + s), \quad r_-(s) = g(a_c - s). \quad (8.66)$$

At either point, the Euclidean clipping direction is parallel to e_1 . Therefore

$$I - e_1 e_1^\top = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \quad (8.67)$$

Since

$$D^2 V(x, 0) = \begin{pmatrix} \lambda - 3bx^2 + 5dx^4 & 0 \\ 0 & \nu \end{pmatrix},$$

the one-step Jacobians along the clipped cycle are

$$DF_{\eta,c}(Y_+(s)) = \begin{pmatrix} 1 & 0 \\ 0 & 1 - \frac{\eta \nu c}{r_+(s)} \end{pmatrix}, \quad (8.68)$$

$$DF_{\eta,c}(Y_-(s)) = \begin{pmatrix} 1 & 0 \\ 0 & 1 - \frac{\eta \nu c}{r_-(s)} \end{pmatrix}. \quad (8.69)$$

Hence the two-step monodromy matrix is

$$M_{\text{clip}}(s) = \begin{pmatrix} 1 & 0 \\ 0 & \left(1 - \frac{\eta \nu c}{r_+(s)}\right) \left(1 - \frac{\eta \nu c}{r_-(s)}\right) \end{pmatrix}. \quad (8.70)$$

Thus the two multipliers are

$$\boxed{\rho_x^{\text{clip}} = 1} \quad (8.71)$$

and

$$\boxed{\rho_y^{\text{clip}}(s) = \left(1 - \frac{\eta \nu c}{r_+(s)}\right) \left(1 - \frac{\eta \nu c}{r_-(s)}\right)}. \quad (8.72)$$

For the symmetric member $s = 0$,

$$\rho_y^{\text{clip}}(0) = \left(1 - \frac{\eta \nu c}{g(a_c)}\right)^2. \quad (8.73)$$

At the switching contact,

$$g(a_c) = c,$$

so the outer limiting transverse multiplier becomes

$$\rho_y^{\text{clip}} = (1 - \eta_{\text{sc}}\nu)^2. \quad (8.74)$$

Since

$$\eta_{\text{sc}}(c) \rightarrow \frac{2}{\lambda}$$

as $c \rightarrow 0^+$ and

$$0 < \nu < \lambda,$$

one has

$$|1 - \eta_{\text{sc}}(c)\nu| < 1$$

for all sufficiently small c .

Hence the clipped family is transversely attracting while remaining exactly neutral along its one-dimensional period-two direction.

8.11 The local transition in the solvable model

The planar model provides an explicit realization of the entire smooth-to-clipped scenario. For $\eta < \eta_f = 2/\lambda$ the origin is locally asymptotically stable; for $\eta_f < \eta < \eta_{\text{sc}}(c)$ there is a unique small stable smooth two-cycle $\{(-a_{\text{sm}}, 0), (a_{\text{sm}}, 0)\}$ whose amplitude grows as

$$a_{\text{sm}} \sim \frac{\lambda}{\sqrt{2b}} \sqrt{\eta - \eta_f}.$$

At $\eta = \eta_{\text{sc}}(c)$ both points reach the clipping boundary simultaneously with

$$a_{\text{sm}} = \frac{\eta_{\text{sc}}c}{2}, \quad \eta_{\text{sc}}(c) = \frac{2}{\lambda} + \frac{2b}{\lambda^4}c^2 + O(c^4).$$

For $\eta > \eta_{\text{sc}}(c)$ sufficiently close to contact, the clipped map possesses a one-parameter family of period-two cycles with exact step constraint $x_+ - x_- = \eta c$, and the smooth critical multiplier $\rho_x^{\text{sm}} < 1$ is replaced by the exact neutral multiplier $\rho_x^{\text{clip}} = 1$. The model thus exhibits the transition from a stable smooth period-two branch, through a simultaneous clipping contact, to a neutral family of clipped two-cycles.

8.12 A concrete parameter choice

To make the preceding formulas explicit, consider the parameter choice

$$\lambda = 2, \quad \nu = \frac{1}{2}, \quad b = 1, \quad d = 1.$$

Then

$$\eta_f = 1, \quad \Gamma_* = 6, \quad \ell_* = 1,$$

and therefore

$$A_* = \sqrt{2}, \quad B_* = 2\sqrt{2}, \quad K_2 = \frac{1}{8}.$$

Hence the smooth period-two branch satisfies $x_{\pm}(\mu) = x_* \pm \sqrt{2\mu}v_* + O(\mu)$, while the first switching threshold obeys

$$\eta_{\text{sc}}(c) = 1 + \frac{1}{8}c^2 + O(c^4),$$

with the next correction computable explicitly:

$$\eta_{\text{sc}}(c) = 1 + \frac{1}{8}c^2 + \frac{1}{64}c^4 + O(c^6).$$

The example thus provides an explicit realization of the general quadratic separation law of Section 6.

9 Conclusion and discussion

We studied the first interaction between a smooth flip bifurcation and the endogenous switching geometry generated by radial gradient clipping. The mechanism operates on two sharply separated scales: clipping is locally invisible near the critical point and becomes dynamically relevant only at finite amplitude.

Local scale. For every fixed $c > 0$ the clipped map $F_{\eta,c}$ coincides with the ordinary gradient map G_η on a neighborhood of a nondegenerate critical point; hence the fixed point, its linearization, its local invariant structures, its center-manifold reduction, and its smooth flip coefficient are independent of the clipping threshold. In particular, for a strict local minimum x_* with simple maximal Hessian eigenvalue λ_* , the first stability threshold remains

$$\eta_f = \frac{2}{\lambda_*}. \quad (9.1)$$

Under the supercriticality condition

$$\Gamma_* = 3T_*[v_*, v_*, H_*^{-1}g_*] - Q_*[v_*^4] > 0, \quad (9.2)$$

the loss of stability creates a stable smooth period-two branch

$$x_\pm(\mu) = x_* \pm A_* \sqrt{\mu} v_* + O(\mu), \quad \mu = \eta - \eta_f, \quad A_*^2 = \frac{3\lambda_*^2}{\Gamma_*}. \quad (9.3)$$

Finite-amplitude scale. Clipping becomes active only when this orbit reaches the switching hypersurface

$$\Sigma_c = \{x : \|\nabla V(x)\|_\diamond = c\}.$$

The gradient structure makes the interaction rigid: every smooth period-two orbit satisfies the exact identity

$$\nabla V(x_+) = -\nabla V(x_-), \quad (9.4)$$

so both points have identical clipping exposure and reach Σ_c simultaneously. Matching the square-root amplitude scale with the threshold produces the principal asymptotic law

$$\boxed{\eta_{\text{sc}}(c) - \eta_f = \frac{\Gamma_*}{3\lambda_*^4 \|v_*\|_\diamond^2} c^2 + O(c^4),} \quad (9.5)$$

which for Euclidean clipping becomes

$$\boxed{\eta_{\text{sc}}(c) = \frac{2}{\lambda_*} + \frac{\Gamma_*}{3\lambda_*^4} c^2 + O(c^4).} \quad (9.6)$$

The even-power expansion reflects an additional symmetry of the gradient problem: in the signed square-root parameter $\varepsilon = \sqrt{\eta - \eta_f}$ the half-difference of the two-cycle is odd while its midpoint is even, which together with (9.4) removes the cubic correction from the contact curve.

Post-contact scale. A clipped period-two orbit cannot mix smooth and clipped points, so any continuation past the first contact enters the clipped regime at both points simultaneously, and every fully clipped period-two orbit obeys the exact step-length constraint

$$\|y_+ - y_-\|_\diamond = \eta c. \quad (9.7)$$

Most importantly, radial clipping creates an unavoidable neutral direction: the monodromy matrix

$$M_c = DF_{\eta,c}(y_-) DF_{\eta,c}(y_+)$$

of a fully clipped period-two orbit satisfies

$$\boxed{1 \in \sigma(M_c).} \quad (9.8)$$

Every fully clipped two-cycle is therefore nonhyperbolic: the radial normalization removes first-order information in the gradient-magnitude direction. The solvable planar model of Section 8 shows that this unit multiplier is not a formal artifact — it corresponds to an actual loss of isolation of the periodic orbit.

9.1 Dynamical interpretation

The results separate two effects that are often conflated. The first is the loss of local stability of the minimum: a smooth gradient-descent phenomenon governed by $D^2V(x_*)$, $D^3V(x_*)$, $D^4V(x_*)$, and unaffected by clipping. The second is the modification of an already finite-amplitude trajectory once it reaches $\|\nabla V\|_\diamond = c$. Clipping does not postpone the first local flip threshold; it introduces a second dynamical scale beyond it, the switching-contact scale $\eta_{\text{sc}}(c) - \eta_f = O(c^2)$. This distinction is useful when interpreting large-step optimization dynamics: linear stability determines when the critical point loses stability, while the clipping threshold determines when the resulting nonlinear oscillation begins to experience a different piecewise-smooth evolution.

9.2 Limitations

The theory focuses on an isolated nondegenerate minimum with a simple maximal Hessian eigenvalue, which isolates a one-dimensional flip mode but excludes the additional neutral directions that arise when the set of minimizers forms a manifold. Second, the post-contact unit multiplier prevents a universal hyperbolic continuation theorem: under the present assumptions alone, the existence, uniqueness, and nonlinear stability of a fully clipped continuation are not determined. Third, the paper treats radial clipping; coordinatewise clipping, adaptive normalization, momentum, and stochastic mini-batch updates generate different switching geometries. Finally, the analysis is local in the bifurcation parameter and does not classify the remote global dynamics after the orbit enters the clipped region.

9.3 Directions for further work

Several extensions appear natural: reducing the nonhyperbolic post-contact dynamics to a higher-order scalar or center-manifold normal form, which the exact unit multiplier suggests may form a new class of degenerate border interactions specific to radial gradient maps; replacing the isolated minimum by a Morse–Bott manifold of minima, combining the -1 flip direction with $+1$ tangent directions; degenerate smooth flips with $\Gamma_* = 0$, where a matching principle $r(\mu) \asymp \mu^\gamma$, $d(c) \asymp c^p$ suggests the general contact scale $\mu_{\text{sc}} \asymp c^{p/\gamma}$; and determining which structures persist for momentum, stochastic gradients, and nonradial clipping.

9.4 Final perspective

Near a nondegenerate minimum, radial gradient clipping is locally invisible: $F_{\eta,c} = G_\eta$, so the smooth map determines when the first oscillatory instability is born. Once the oscillation reaches finite amplitude, the same potential generates an endogenous switching boundary at the explicit scale

$$\eta_{\text{sc}}(c) - \eta_f \sim \frac{\Gamma_*}{3\lambda_*^4 \|v_*\|_\diamond^2} c^2.$$

At that point the dynamics cease to be governed solely by the smooth flip normal form: radial clipping removes the gradient-magnitude degree of freedom and forces the post-contact period-two dynamics to carry an exact neutral multiplier. The clipped gradient map thus exhibits a sharp transition from smooth bifurcation theory to intrinsically nonhyperbolic switching dynamics, with the transition scale determined explicitly by the local geometry of the underlying potential.

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The authors declare that they have no conflict of interest.

Author contributions

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